INSTRUCTIONS

GEK-36416A

3S7932LA202

POWER SYSTEM STABILIZER

Note: Includes Frequency Transducer with Input 60 Hz Bandpass Filter.

GENERAL ELECTRIC

GENERAL ELECTRIC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE: UNDER-VOLTAGE OVERVOLTAGE DETECTION
FIRST MADE FOR
ELEMENTARY DIAGRAM

REFERENCE AMPLIFIER — COMPARATOR — LOGIC CIRCUIT

+15V −15V
IN2 10K D5R D3D D4D D2C .22 100K D10R
IN1 +30V D1TP D1R 402 806 D2R INPUT VOLTAGE D1R 100 D4R 10K D1C 5 D3R 200
GRD2
D2D D1D D1A TRIM R TRIM *
D26R 4.99K
10K D6R
D3C .1 D4C 47 D5C .1 D6C 47
REF. D3TP −15V +15V D4TP
−15V 680 D7R 221 D8R 6.2V IN825 D1Z D2P 500 D2TP −5V D9R 1.62K
D10R 3.9K D11R 10K D13R 10K D14R 3.9K D12R
5.6K D16R
D8C .1 D9C .1
D5D D6D D7C 2.2
D2A
5.1K D17R NEG. LIMIT D3P 2K 0
220 D18R D4P 500 NEG. REF. 820 D19R +15V D5TP D2Z 6.2V
5.1K D28R
D8TP
D9TP 4.7V
+15V 150Ω D24R D4Z 10K D25R D13C 33 D9D
D10TP ONE SHOT D1L D11TP
D2L
D13TP D12TP
D3L
D14C 220pF
D27R 220 OUTPUT OUT2 OUT2 (TEST)
OUT1 5.6K D20R
D11C .1 D12C .1
D8D D7D D10C 22
D3A
5.1K D21R POS. LIMIT D5P 2K 0
220 D22R D6P 500 POS. REF. 820 D23R −15V D6TP −5V D3Z 6.2V
D29R 5.1K
D7TP

NOTES:
⊗ - MOUNTED ON FACEPLATE
RESISTOR VALUES ARE IN OHMS (Ω) UNLESS OTHERWISE SPECIFIED.
CAPACITOR VALUES ARE IN MFD. UNLESS OTHERWISE SPECIFIED.
* - RESISTOR VALUE WILL VARY WITH EACH UNIT.
⊠ BOARD MOUNTED TEST POINT.

Figure 7, Sheet 9

REVISIONS	PRINTS TO
1 Feb 6, 1974	11 769

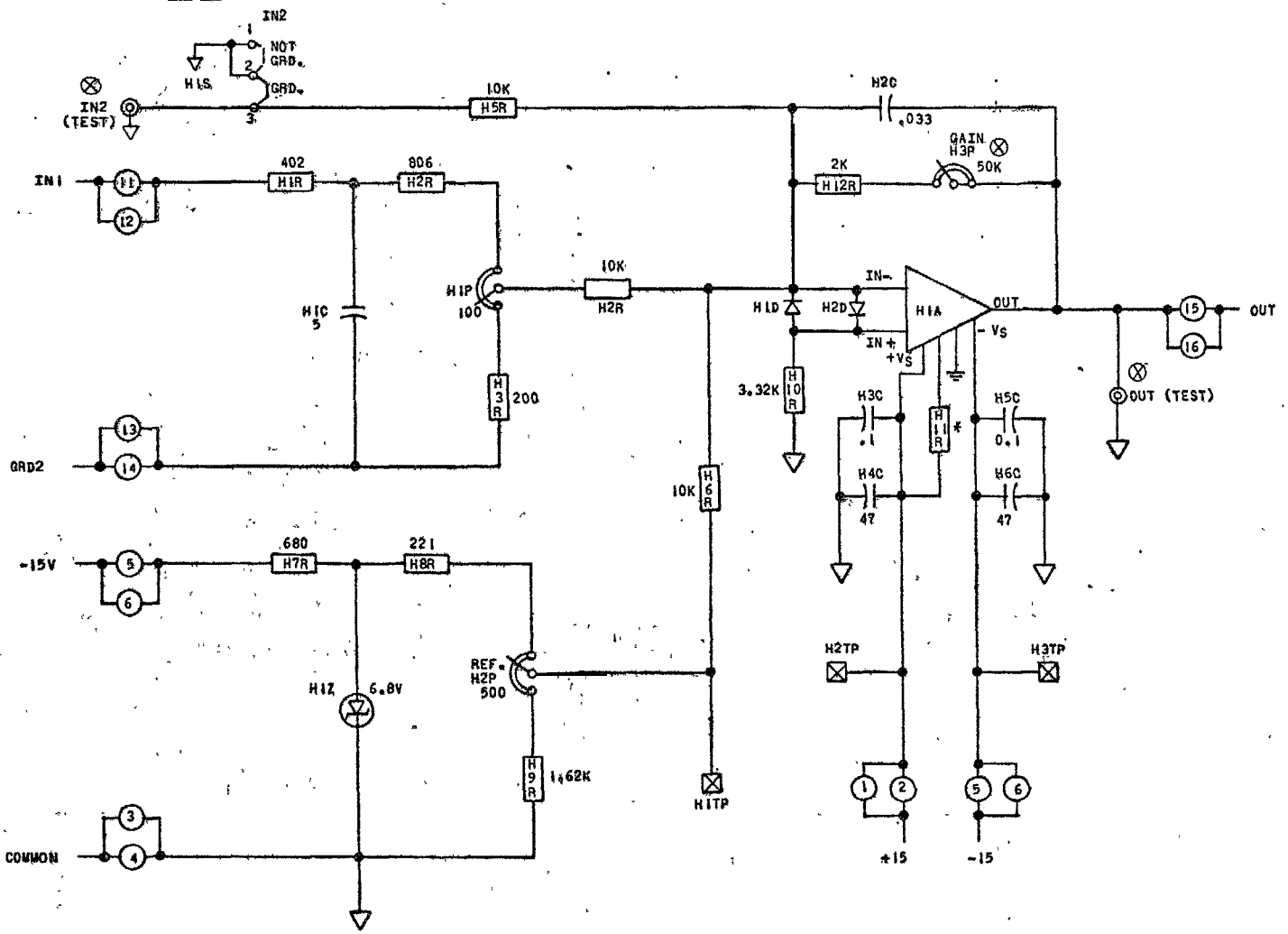
MADE BY 4/18/73 | APPROVALS T. Smith | DRIVE SYSTEMS PROJ. DEPT. WAYNESBORO, VA. LOCATION | 44C306566 CONT ON SHEET | SH NO 1
ISSUED D. Mayer May 21, 1973

GENERAL ELECTRIC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
VOLTAGE DEVIATION SENSING
POWER SYSTEM STABILIZER
FIRST MADE FOR 3S7932MA324G1, LA202G1
ELEMENTARY DIAGRAM

44C306564 CONT ON SHEET SH NO



NOTES:

⊗ = MOUNTED ON FACEPLATE

RESISTOR VALUES ARE IN Ω AND ARE 1/2 WATT.
CAPACITOR VALUES ARE IN MFD.

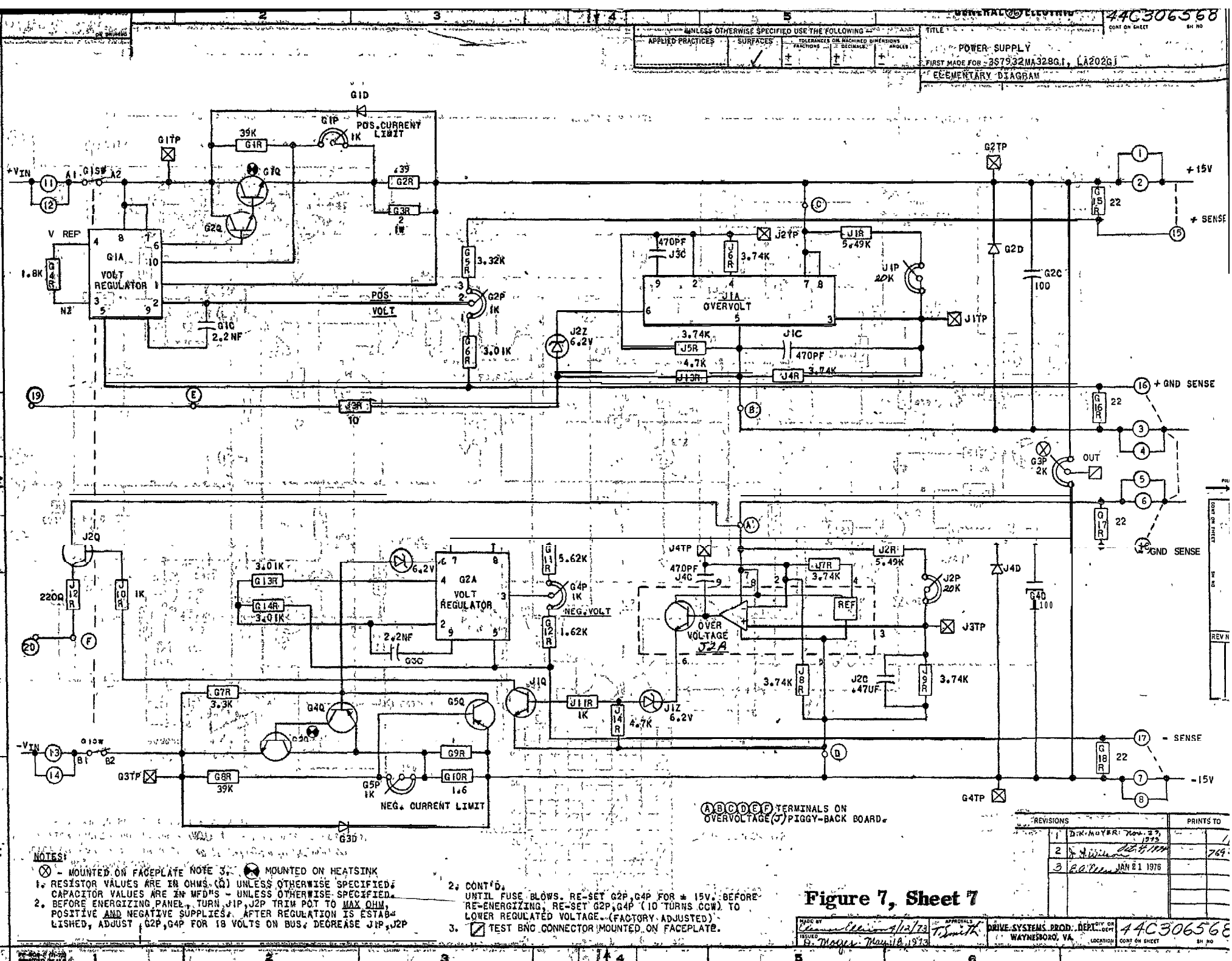
Figure 7, Sheet 8

44C306564 CONT ON SHEET SH NO
REV NO

REVISIONS | PRINTS TO

MADE BY LEV MATER 9/7/73 | APPROVALS T. Smith | DRIVE SYSTEMS PROD. DEPT. V OR DEPT WAYNESBORO, VA. LOCATION | 44C306564 CONT ON SHEET SH NO

FF-300-7 IN-70 PRINTED IN U.S.A.



NOTES:

⊗ - MOUNTED ON FACEPLATE NOTE 3. ⊠ MOUNTED ON HEATSINK

1. RESISTOR VALUES ARE IN OHMS (Ω) UNLESS OTHERWISE SPECIFIED. CAPACITOR VALUES ARE IN MFD'S - UNLESS OTHERWISE SPECIFIED.
2. BEFORE ENERGIZING PANEL, TURN J1P, J2P TRIM POT TO MAX OHM, POSITIVE AND NEGATIVE SUPPLIES. AFTER REGULATION IS ESTABLISHED, ADJUST G2P, G4P FOR 18 VOLTS ON BUS. DECREASE J1P, J2P
2. CONT'D. UNTIL FUSE BLOWS. RE-SET G2P, G4P FOR ± 15V. BEFORE RE-ENERGIZING, RE-SET G2P, G4P (10 TURNS CCW) TO LOWER REGULATED VOLTAGE. (FACTORY ADJUSTED)
3. ☒ TEST BNC CONNECTOR MOUNTED ON FACEPLATE.

Figure 7, Sheet 7

REVISIONS			PRINTS TO
1	DK-MOYER	Nov. 29, 1973	
2		Dec 4 1974	769
3	R.A. Reese	JAN 21 1976	

MADE BY	APPROVALS	DRIVE SYSTEMS PROD. DEPT.	DIV. OR DEPT	44C306568
ISSUED R. Mayes May 18, 1973	T. Smith 4/2/73	WAYNESBORO, VA.	LOCATION	CONT ON SHEET / SH NO

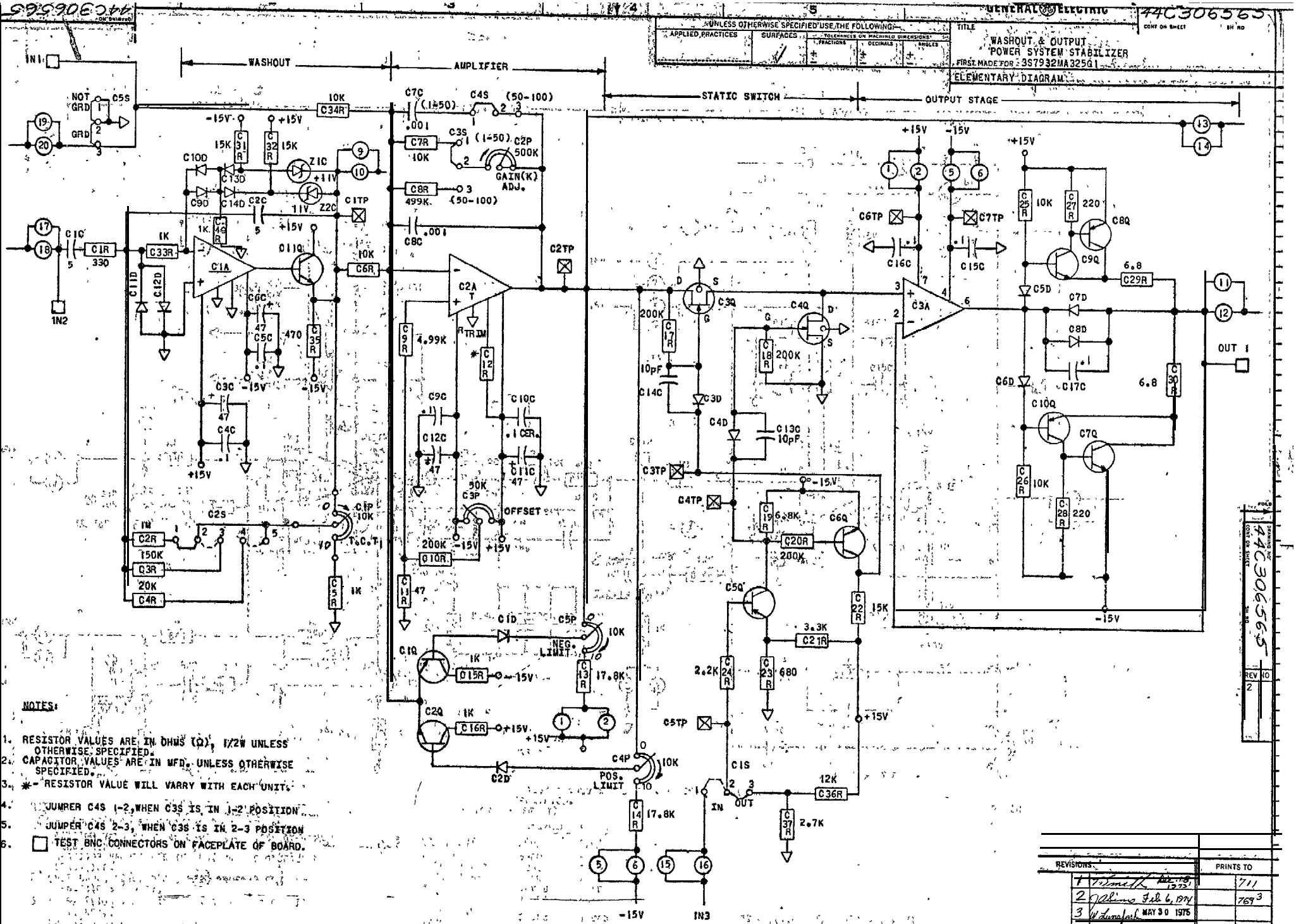


Figure 7, Sheet 4

REVISIONS			PRINTS TO
1	T. Smith	Dec 18 1973	711
2	J. Sims	Feb 6, 1974	769 3
3	W. Lunsford	MAY 30 1975	
4	E.A. Elmore	JAN 21 1976	
5	E.A. Elmore	SEP 16 1976	
6	D. Sims	NOV 16 1976	

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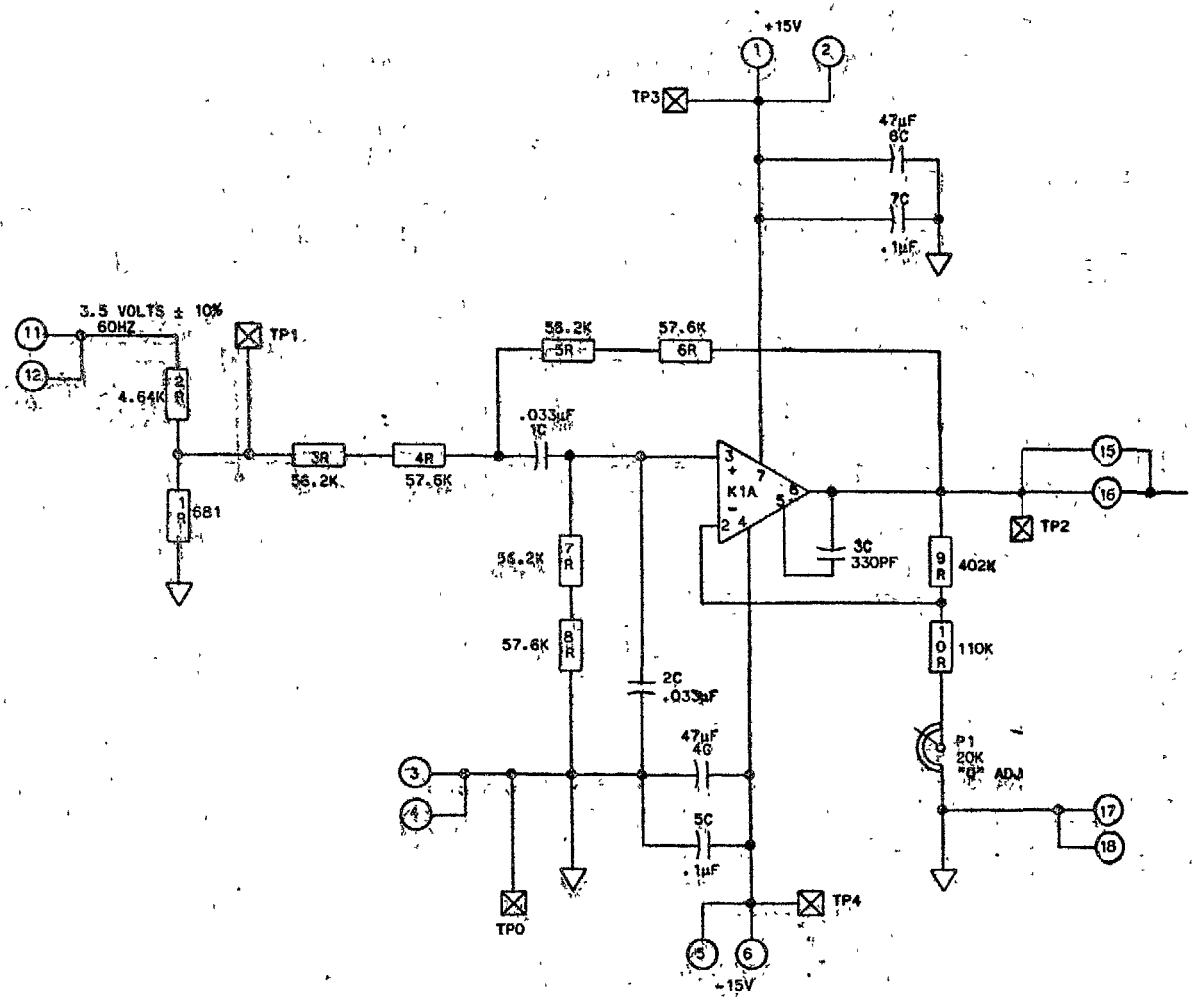


Figure 7, Sheet 1

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING –
APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE POWER SYSTEM STABILIZER
FIRST MADE FOR 3S7932LA202G2
FUNCTIONAL ELEMENTARY

NOT FURNISHED BY GE WAYNESBORO
MOUNTED ON CUSTOMER CONTROL BOARD
CUSTOMER TERMINAL BOARD
INTERNAL TERMINAL BOARD POINT
REDUNDANT PC BOARD PINS
TERMINAL BOARD POINTS ON BACK OF PC BOARD RACK
INDICATOR LIGHTS
LIGHT EMITTING DIODE (LED)
RELAY OR CONTACTOR COIL
RELAY CONTACT
RESISTOR
POTENTIOMETER
CAPACITOR
INDUCTANCE, TRANSFORMER WINDING
FUSE
DISCONNECT SWITCH
RHEOSTAT, ADJUSTABLE RESISTOR
RECTIFIER (DIODE)
THYRISTOR (SCR)
ZENER DIODE
OPERATIONAL AMPLIFIER (OP AMP)
TRANSISTOR
PUSHBUTTON (N.O.)
PUSHBUTTON (N.C.)
CIRCUIT COMMON
TWISTED SHIELDED LEADS
AND CIRCUIT
PCB MOUNTED SELECTOR SWITCH
METAL OXIDE VARISTOR (M.O.V.)

CIRCUIT	SHEET
60HZ BAND PASS FILTER	6
CEILING SSENSING	3
FREQUENCY TRANSDUCER	2
POWER SUPPLY	
+15 VOLT	5
24 VOLT	4
RELAY	4
SIGNAL CONDITIONING	3
WASHOUT AND OUTPUT	3

FOLD

CONT ON SHEET | SH NO. | REV NO.

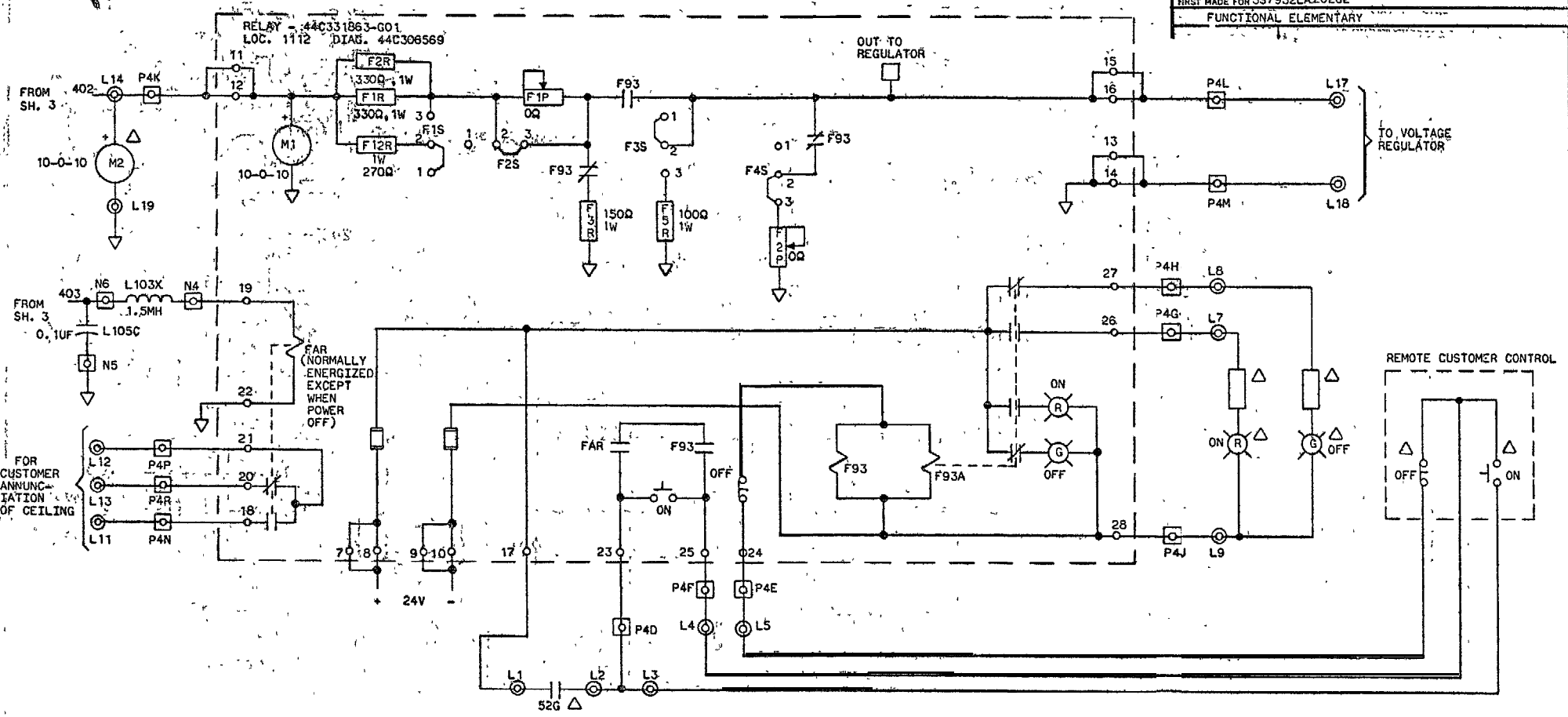
REVISIONS | PRINTS TO 711

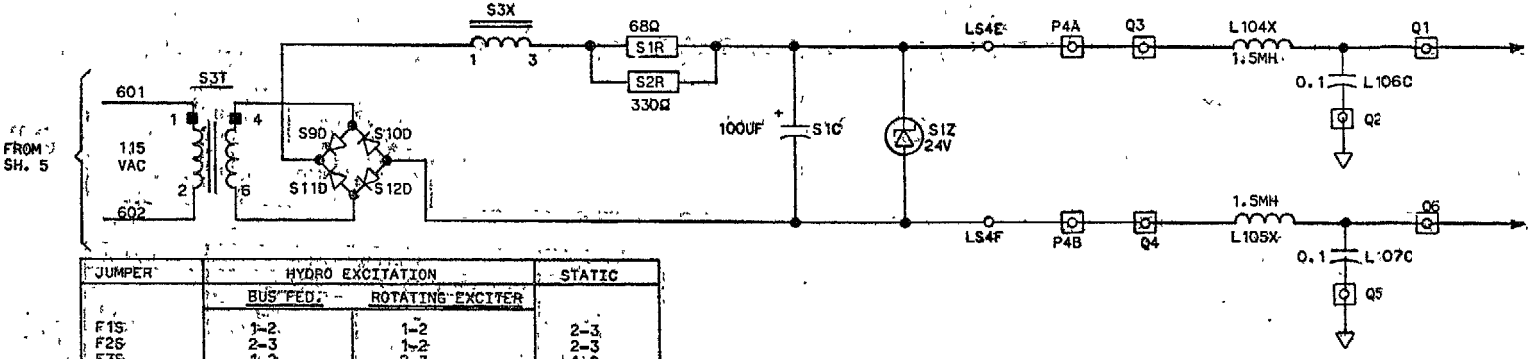
MADE BY H J Shutt Oct 6, 1976 ISSUED OCT 25 1976
APPROVALS RO Rawson Kpl
DRIVE SYSTEMS PROD. DEPT. DIV OR DEPT. WAYNESBORO, VA.
44C321824 CONT ON SHEET FNL SH NO. 6

Figure 6, Sheet 6

Figure 6, Sheet 5

TITLE
POWER SYSTEM STABILIZER
FIRST MADE FOR 3S7932LA202G2
FUNCTIONAL ELEMENTARY





JUMPER	HYDRO EXCITATION		STATIC
	BUS FED	ROTATING EXCITER	
F1S	1-2	1-2	2-3
F2S	2-3	1-2	2-3
F3S	1-2	2-3	1-2
F4S	2-3	2-3	2-3

POTENTIOMETER			
F1P	0	225	0
F2P	0	390	102

Figure 6, Sheet 4

REVISIONS
NOV 10 1976
MAY 4 1977

PRINTS TO 711

DRAWN BY R.E. Hayden 10-14-76 ISSUED OCT 25 1976
APPROVALS
DRIVE SYSTEMS PROD. DEPT'Y OR DEPT. WAYNESBORO, VA.
44C321824 CONT'D SHEET 5 SH NO 4

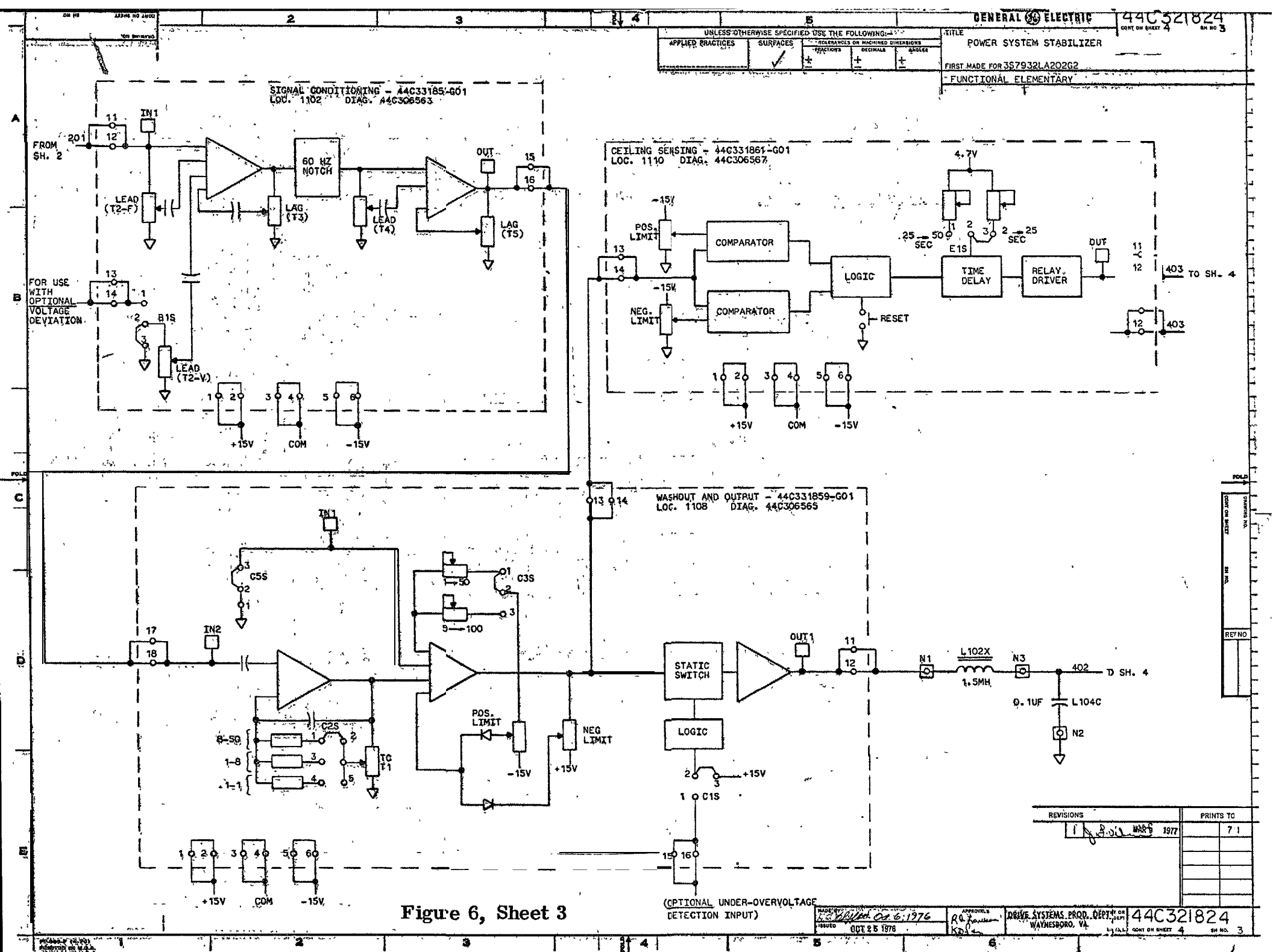
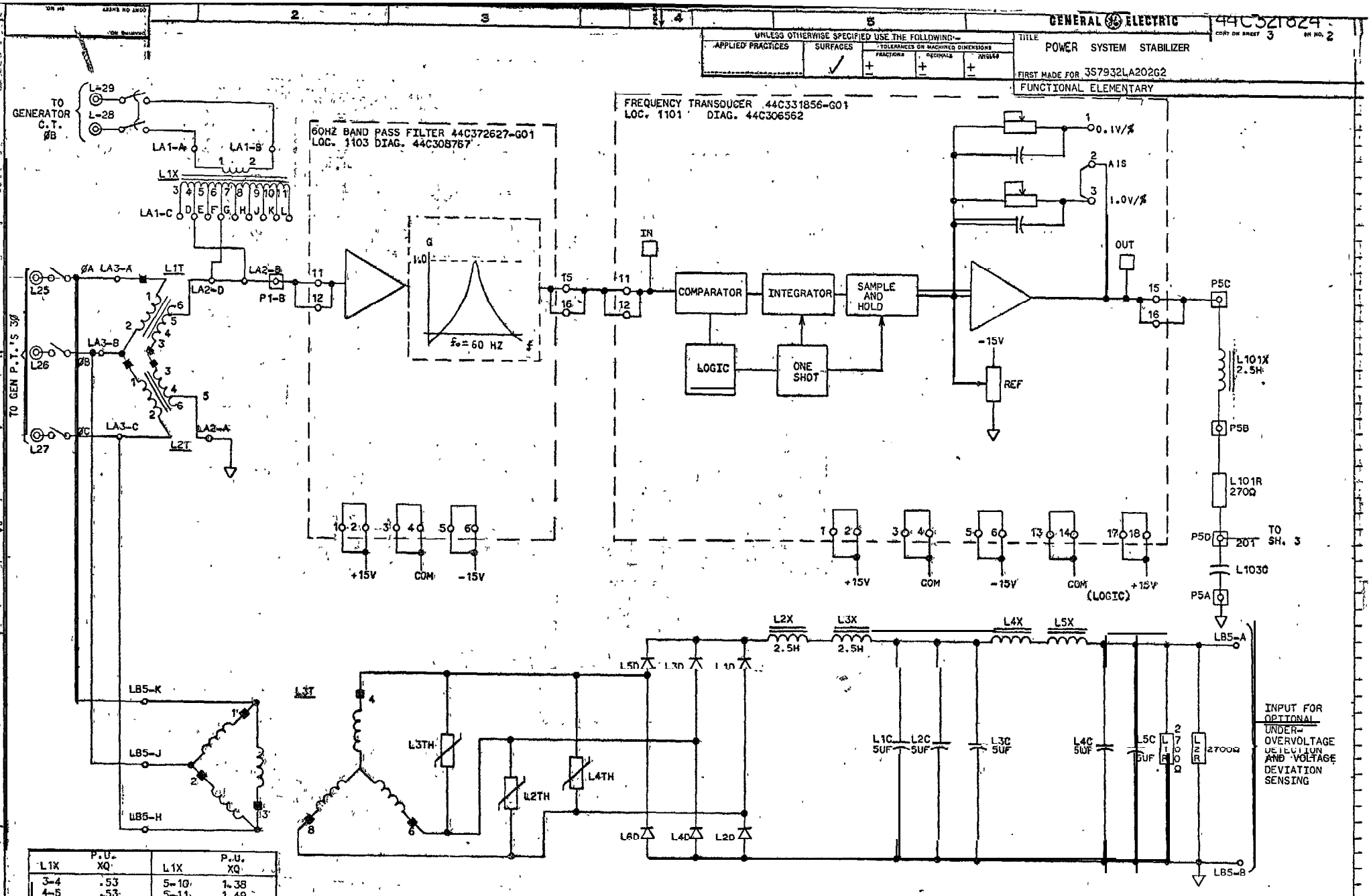


Figure 6, Sheet 3



L1X	P.U. XQ	L1X	P.U. XQ
3-4	.53	5-10	1.38
4-5	.53	5-11	1.49
5-6	.53	4-7	1.6
6-7	.53	4-8	1.71
6-8	.64	4-9	1.81
6-9	.74	4-10	1.92
6-10	.85	4-11	2.03
3-5	1.06	3-7	2.13
4-6	1.06	3-8	2.23
5-7	1.06	3-9	2.34
5-8	1.17	3-10	2.44
5-9	1.27	3-11	2.55
5-10	1.38	6-11	0.96

Figure 6, Sheet 2

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE POWER SYSTEM STABILIZER

FIRST MADE FOR 3S7932LA202G2

FUNCTIONAL ELEMENTARY

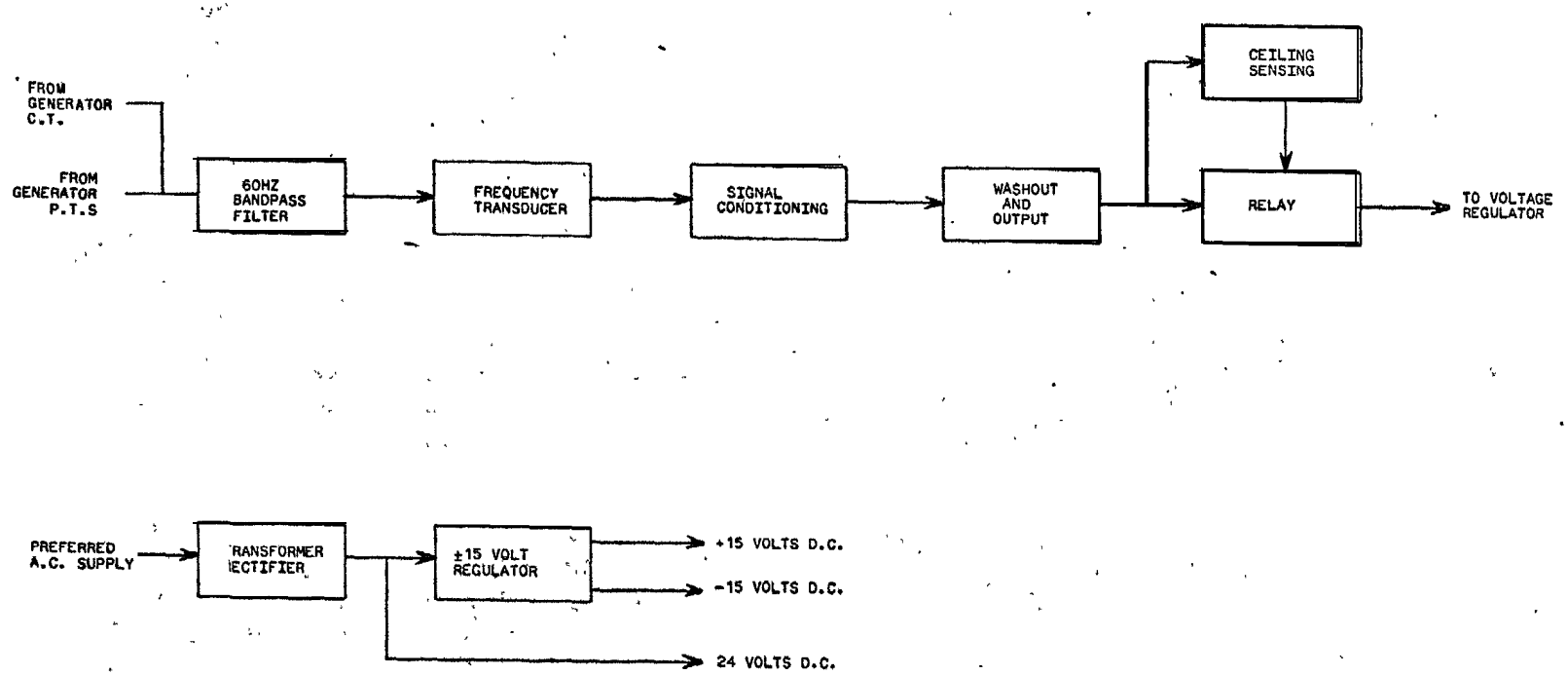


Figure 6, Sheet 1

REVISIONS

PRINTS TO 711

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INTRODUCTION

The Power System Stabilizer (PSS) is designed to introduce a supplementary control signal into continuously acting voltage regulator equipment. The equipment will supplement the conventional voltage regulator control with a control function derived so that, under deviations of generator frequency, excitation becomes controlled as a composite function of voltage and frequency. Under certain conditions, the composite function has been shown to be an aid to power system stability.

The PSS receives a signal from generator terminal voltage (j-phase). This signal, which is frequency modulated, is passed through a bandpass f i 1 ter to remove excessive power system noise and a frequency transducer which then has an output frequency equal to the modulation frequency and an ampl itude proportional to the deviation in frequency. This signal is then passed through two lead-lag stages in order to obtain the proper phase shift characteristic, a washout stage to eliminate the effect of component drift and slow system frequency variations, a gain stage to obtain maximum control action and an output stage which interfaces the PSS with continuously acting voltage regulating equipment.

The Power System Stabilizer includes the functions of input transducer, signal conditioning, limiting, and output circuitry. Figure 6 is a typical functional elementary diagram while Figure 7 is composed of the printed circuit board elementary diagrams.

The Power System Stabilizer will apply the following transfer function to modify the output of the speed transducer for insertion into the voltage regulator:

$$G(s) = \frac{K\,(T_1S)\,(1 + T_2S)\,(1 + T_4S)}{(1 + T_1S)\,(1 + T_3S)\,(1 + T_5S)}$$

Adjustments are provided to vary the parameters of the PSS so as to permit compensation for variations in the characteristics of the regulator, excitation system, generator and power system to which it is applied. These adjustments. can be made by calibrated dial settings which will be accurate to within 10% of the desired parameter value. The adjustments are capable of modifying the transfer function parameters through the fol lowing typical ranges :

K = 0.1 to 100

T_1 = 0.1 to 50 seconds

T2 = 0.2 to 1.5 seconds

T_3 = 0.02 to 0.1 seconds

T_4 = 0.2 to I.5 seconds

T_5 = 0.02 to 0.1 seconds

RECEIVING AND HANDLING

The Power System Stabilizer is normally supplied mounted in either the regulator or exciter cubicle. The instruction manual for the particular excitation system used should be consulted for receiving and handling instructions. If the PSS is shipped separately (as in the case of a retrofit system), then immediately upon receipt the equipment should be carefully unpacked to avoid damage. It should be examined for any damage that might have been sustained in transit. If injury or rough handling is evident, a damage claim should be filed immediately with the transportation company and the nearest General Electric sales office should be notified promptly.

STORAGE

If the equipment is not to be used as soon as it is unpacked, it should be stored in a clean, dry place and protected from accidental damage. Particular care should be exercised to avoid storing the equipment in locations where construction work is in progress. If equipment is to be stored for more than three months, it should be kept heated to at least 5°C above ambient temperature to prevent condensation.

ABBREVIATIONS

The following abbreviations are used in this instruction manual:

ac	Alternating Current
Amp	Amperes
Db	Decibels
dc	Direct Current
FET	Field Effect Transistor
Hz	Hertz
IC	integrated Circuit
Ma	Milliamp
Mv	Millivolt
PP	Peak-to-Peak
PSS	Power System Stabilizer
SCR	Silicon Controlled Rectifier
VAC	Alternating Current Voltage

ABBREVIATIONS (continued)

VDC	Direct Current Voltage
GRD	Ground

(DESCRIPTION!

The Power System Stabilizer consists of:

3S7932LA202G2	Panel

The Power System Stabilizer is normally mounted in either the regulator cubicle or exciter cubicle and contains the following:

Number	Name	Quantity
44C372627-G01	60 Hz Bandpass Filter Board	1
44C331856-G01	Frequency Transducer Board	1
44C331857-G01	Signal Conditioning Board	1
44C331859-G01	Washout and Output Board	1
44C331863-G01	Relay Board	1
44C331867-G01	Ceiling Sensing Board	1
44C331862-G01	Power Supply Board	1
3S7932MA321G1	Internal Frequency Panel	1
3S7932MA321G2	Terminal Voltage Sensing Panel	1
3S7932MA321G3	± 36 Volt Power Supply Panel	1
3S7932MA321G4	24 Volts Power Supply Panel	1
44B331785-G01	Extender Board	1

INSTALLATION

LOCATION AND CONNECTIONS

The Power System Stabilizer is normally furnished as an integral part of the excitation and regulation equipment. Refer to the drawings furnished with the excitation system and the instruction manual for the excitation system for location and connection instructions.

INSTALLATION (continued)

INITIAL CHECKS

1. Wire check all external connections *to* the PSS. Be sure that wire size and type are adequate for the expected signal levels. Refer to the connection diagram furnished with the excitation equipment.

CAUTION

DO NOT HI-POT OR MEGGER ANY CIRCUIT AS THIS MAY RESULT IN DAMAGE TO COMPONENTS. ANY WIRING OR CIRCUITS TO BE HI-POTTED OR MEGGERED MUST FIRST BE DISCONNECTED FROM THE PSS.

2. Insure that all printed circuit boards are fully inserted.

3. insure that all shielded cables have properly terminated shields.

OPERATION]

CAUTION

MISOPERATION OF THE POWER- SYSTEM STABILIZER WHILE CONNECTED TO THE EXCITATION SYSTEM MAY RESULT IN DAMAGE TO THE GENERATOR. DEVIATIONS FROM THE FOLLOWING OPERATIONAL CHECKS AND PROCEDURES SHOULD BE CONSIDERED IN LIGHT OF THIS AND SHOULD BE DISCUSSED WITH THE MANUFACTURER PRIOR TO INITIATION.

INITIAL OPERATION

CAUTION

THE POWER SYSTEM STABILIZER MUST BE OUT OF SERVICE OR THE REGULATOR IN THE MANUAL MODE FOR ALL TESTS THROUGH PARAGRAPH 10 (EXCEPT PARAGRAPH 2 AS NOTED). FAILURE TO DO THIS MAY RESULT IN DAMAGE TO THE GENERATOR.

1. Turn on ± 15 VDC power using the toggle switch mounted on the faceplate of the Power Supply board (LOC 1111). Check that power supply voltages are correct.

   Adjustment of the ± 15 VDC supply can be made with board-mounted rheostats. (Refer to theory of operation section.)

OPERATION (continued)

INITIAL OPERATION (continued)

CAUTION

ON SOME EXCITATION EQUIPMENT, PULLING THE RELAY BOARD INTRODUCES A STEP INTO THE REGULATOR. THIS CAN RESULT IN DAMAGE TO THE GENERATOR. INSURE THAT THE REGULATOR IS IN THE MANUAL MODE OF OPERATION BEFORE PULLING THE RELAY BOARD.

2. If the generator is in the automatic mode of operation, transfer to manual. Pul? the Relay board (LOC 1112). The Relay board is designed to interface with many excitation systems. Board-mounted shorting clips and rheostats are used to adapt the relay board to the particular excitation system. (Refer to the drawings furnished with the equipment for the settings of F1S, F2S, F3S, F4S, F1P and F2P.) Check settings and readjust if necessary. Insert Relay board. Pu? 1 Washout and Output board (LOC 1108). t i n g cl ip C5S on terminals 1, 2 (not GRD) . Insert Extender board in LOC 1108.

CAUTION

THE REGULATOR MUST BE IN THE MANUAL MODE OF OPERATION FOR THIS TEST. FAILURE TO PLACE THE REGULATOR IN MANUAL MAY RESULT IN DAMAGE TO THE GENERATOR.

Insert a signal from pins 11, 12 to pins 3, 4 of the Extender board. This signal can be a dc signal or an oscillating signal (peak value should be less than 10 volts). With the PSS off (as evidenced by the "OFF" pushbutton being lighted) no voltage should exist at the test jack labeled "Out to Regulator" mounted on the Relay board. Pressing the "ON" pushbutton should cause the voltage to be present but attenuated for insertion into the appropriate regulator. If the "ON" pushbutton will not stay energized, then press the reset pushbutton mounted on the faceplate of the Ceiling Sensing. boa rd. If the "ON" pushbutton wi 11 sti 11 not stay energized, then check that the permissive contact is closed, usually 52G line breaker (see sheet 4 of Figure 6). If the unit is off line, then this permissive contact must be jumpered in order to establish that the output relay is working properly. Press the "OFF" pushbutton. The voltage at the test jack should be zero and the "OFF" pushbutton should be 1 ighted. Remove Extender board. I nsert Washout and Output board in LOC 1108. On Ceiling Sensing board (LOC 1110), put the.de?ay dial on zero, the positive limit dial on 500 (5 turns) and the negative limit dial on 500. Press the reset pushbutton on the Ceiling Sensing board.

Adjust the gain for 1.0 by turning the dial marked "Gain K" on the Washout and Output board to zero. Connect a variable source of dc voltage to "IN 1"

OPERATION (continued)

INITIAL OPERATION (continued)

of the Washout and Output board and set the voltage to some value less than 5 volts. Press the "ON" pushbutton on the Relay board. A voltage should exist at the test jack on the Relay board labeled "Out to Regulator". Increase the voltage at "IN1" of the Washout and Output board to ≈6 volts. After approximately 2.5 seconds, the FAR relay should drop out causing the "OFF" pushbutton to be lighted. Try to reinsert the PSS by pressing the "ON" pushbutton. The PSS should not latch in the on position. Remove the voltage from IN1. Press the reset pushbutton on the Ceiling Sensing board. Put the PSS in service (with the generator control in manual) by pressing the "ON" pushbutton. The PSS should latch in the on position and the "ON" pushbutton should be lighted. Remove the PSS from service by pressing the "OFF" pushbutton. The "OFF" pushbutton should be lighted. Some installations may have customer controls for the PSS mounted in the control room. Check that any external customer controls can be used to insert and remove the PSS. Leave the PSS in the off position.

CAUTION

THE FOLLOWING GAIN TEST MUST BE PERFORMED WITH THE PSS IN SERVICE AND THE REGULATOR IN THE AUTOMATIC MODE. INSURE THAT THE SHORTING CLIPS AND RHEOSTAT SETTINGS ON THE RELAY BOARD ARE CORRECT FOR THIS EQUIPMENT. INSURE THAT THE FREQUENCY TRANSDUCER BOARD IS PULLED. FAILURE TO DO THI S MAY RESULT IN DAMAGE TO THE GENERATOR.

The Relay board contains the necessary interface circuitry so that a one volt change in PSS output wi 11 cause a one percent change in generator open-circuit terminal voltage. With the PSS off, a small step of voltage (≈ 1 VDC) should be inserted into IN1 of the Washout and Output board. The regulator should be in the automatic mode of operation with the generator preferably off line. The test may be satisfactorily performed on 1 ine if the reactive current compensator (if any) is bypassed and if the voltage regulator of any other generator units paralleled at terminal voltage is temporarily turned to manual control. The gain is calculated by dividing the percent change in generator terminal voltage by the step in voltage applied to IN1. The gain should be (-) 1%/volt ± 0.1. The minus sign (terminal voltage decreases for a positive input voltage) arises because the signal is inserted into IN1, which bypasses the washout stage. F1P (on the Relay board) can be used to adjust the gain if necessary.

An attempt has been made to interface with the voltage regulator in such a way that placing the PSS into and out of service will not cause a bump. On some equipments, F2P on the Relay board has been furnished for this purpose. With the voltage regulator in manual (or test if a test position is available on the control switch) pull the Frequency Transducer (LOC 1101). Place the PSS in service using the pushbuttons on the Relay board. Note whether the

OPERATION (continued)

INITIAL OPERATION (continued)

ac regulator output voltmeter or transfer voltmeter changes when the PSS is put in service. Use F2P on the Relay board *to* minimize the change in ac regulator output voltage when the PSS is inserted. Remove the voltage at IN1 and remove the PSS from service.;

3. Check out -the Internal Frequency panel. This panel is mounted below the printed circuit board rack and contains voltage transformers which accept 3-phase voltage (115 VAC) and provide an output-voltage proportional to phase B 1 ine-to-neutral voltage. Check that three-phase voltage having the correct amplitude and phase sequence is connected to L25, L26 and L27 (ØA, ØB and ØC, respectively). With 115 VAC on the primary of L1T and L2T and the isolating switches closed, the voltage measured from LAZ-A to LAZ-D should be 3.5 VAC, ± 0.5 VAC. Secondary taps are provided on L1T and L2T to compensate for a ± 5 volt variation in primary voltage. For minimum noise, a shielded twisted cable should be used from the voltage source to terminals L25, L26 and L27 (ground shield at the PSS panel only). The Internal Frequency panel can be used to generate a voltage having so-called "internal frequency" by vectorially adding a phase shifted current signal to the voltage signal. This is done using a reactor and a current signal proportional to phase B generator line current. The voltage developed is then V = VT + j IX. The value of the reactance is normally chosen'to be equal to X_q in per unit.

If it is desi red to use the "internal frequency" feature, a shorting jumper from LAZ-D to LAZ-B must be removed and the shorting switch in the primary of LIX closed. With rated voltage on the generator and supplying unity power factor load current, a dual trace scope (or 2 channel brush recorder) can be used to determine if the voltage and current signal have the proper phasing. For a unity power factor load, the voltage from LAZ-D to LAZ-B should lead the voltage from LAZ-A to LA2-D by 90 degrees. For underexcited operation (leading power factor), the voltage will lead by more than 90 degrees and for overexcited operation (lagging) the voltage will lead by less than 90 degrees. If a two channel oscilloscope is used, connect the common of the oscilloscope to LAZ-D with one probe on LAZ-B and the second probe on LAZ-A. This will cause an inversion of the voltage signal so use the inversion feature of the oscilloscope to invert channel two. The signal on channel one (proportional to line current) will lead the voltage by 90 degrees (at unity power factor).

4. Check out the 60 Hz Bandpass Filter board (LOC 1103). The 60 Hz Bandpass Filter is used to remove excessive power system noise before insertion into the Frequency Transducer.' The only adjustment on this board is the "Q" of the filter (P1). Using an oscillator, insert a signal from LAZ-B to common. The signal should be approximately 3.5 VAC at 60 Hz. The gain through the 60 Hz Bandpass- Filter should be 1.2 + 0.2. Determine the upper frequency at which the response is down by 3Db (0.707 times the value at 60 Hz). Determine the lower 3Db frequency by the same procedure. The "Q" of the filter is defined as $Q = fo/(f_2-f_1)$ where fo is the center frequency (60 Hz), f_2 is the upper 3Db frequency and f_1 is the lower 3Db frequency. The "Q"

OPERATION (continued)

INITIAL OPERATION (continued)

has been factory adjusted to be 3.0. If the measurements indicate that the "Q" deviates by more than 10% from the desired value, adjustment should be made using P1 (this is an adjustment that can be made through a hole in the faceplate). It should be noted that *an* adjustment in "Q" affects the gain at center frequency so that a new center frequency gain should be used in determining the frequency at which the gain is reduced by 3Db.

5. Check out the Frequency Transducer board (LOC 1101). Pull the 60 Hz Bandpass Filter (LOC 1103). Check that A1S shorting clip on the Frequency Transducer board is on terminals 2 and 3. Using an oscillator, insert a signal into IN test jack located on the faceplate. Adjust the signal for 60 Hz having an amplitude of approximately 3.5 VAC. The output of the frequency transducer should be 0 ± 0.1 VDC. Don't make any adjustments in this null unless the output voltage deviates from the limits given and the frequency of the osci l lator is accurately known. Adjustment can be made later with the generator on line at rated voltage by using an Extender board. Change the frequency by ± 0.6 Hz (± 1%) and *note* the output voltage.. The gain should be 1 volt/percent so the total change in output should be +l - (-1) = 2. Do not adjust the gain unless it deviates by more than 10% from 1 volt/%. Change switch A1S *to* 1, 2. The gain should now be 0.1 volt/% ± 10%. Change switch A1S back to 2, 3. The frequency transducer output should be positive for an increase in frequency and negative for a decrease in frequency (assuming the output is at zero volts at 60 Hz). A "jog" pushbutton is provided which inserts a step into the PSS for test purposes. With 60 Hz on the input to the frequency transducer, press the "jog" pushbut ton. The output vol tage should decrease by approximately 5Mv. Release the pushbutton and the voltage should return to the original value.

6. If it becomes necessary to check the frequency response of the PSS, set Signal Conditioning board (LOC 1102) for lead time constants of 0.5 seconds and lag time constants of 0.05 seconds. This can be accomplished by the following adjustments:

TCT2V = 0 turns

TCT2F = 2.7 turns

TCT3 = 4.0 turns

TCT4 = 2.7 turns

TCT5 = 4.0 turns.

On the Washout and Output board (LOC 1108) set the washout time constant for 10 seconds and the gain for 1.0. This can be accomplished by placing the shorting clip on C2S on terminals 1, 2, TCT1 dial on 5.4 turns and the gain control at zero. The twin tee notch filter on the Signal Conditioning board should be adjusted for the proper frequency (60 Hz).

OPERATION (continued)

INITIAL OPERATION (continued)

CAUTION

THE PSS MUST BE OUT OF SERVICE FOR THE FREQUENCY RESPONSE TESTS. FAILURE TO DO THIS MAY RESULT IN DAMAGE TO THE GENERATOR.

Pull the Frequency Transducer board (LOC 1101). Insert the Extender board in place of the Frequency Transducer board. Connect an oscillator to terminal 15 or 16 and terminal 3 or 4 of the Extender board. Measure the frequency response of the PSS from 0.1 Hz to 100 Hz measuring oscillator output and output test connector on the Washout and Output board. The oscillator output level should be held at 100 Mv zero to peak (ZOO Mv PP) to avoid saturation of the amplifiers. Compare gain and phase to values shown in Figure 1.

7. Check out the Ceiling Sensing board (LOC 1110). Pull the Ceiling Sensing board and verify that *the* shorting clip on E1S is between 1 and 2. Reinsert the board. Set both the positive limit (E2P) and negative 1 imit (E3P) for 500 (5.0 turns). Set the time delay on the Ceiling Sensing board for minimum (0). Insert a variable dc voltage into IN1 of the Washout and Output board. Place the regulator in the manual mode of operation.

CAUTION

THE REGULATOR MUST 'BE IN THE MANUAL MODE OF OPERATION FOR THIS TEST. FAILURE TO DO THIS COULD RESULT IN DAMAGE TO THE GENERATOR.

Place the PSS in service by pressing the "ON" pushbutton on the Relay board. Slowly increase the voltage until the PSS trips out of service as evidenced by the "OFF" pushbutton being lighted. The voltage at Out 1 on the Washout and Output board should be 5 volts ± 0.25 volts. Remove the voltage. Pressing the "ON" pushbutton on the Relay board should have no effect. Press the reset pushbutton on the Ceiling Sensing board. Pressing the "ON" pushbutton on the Relay board should now cause the PSS to be placed in service. Reverse the voltage into IN1 of the Washout and Output board. The same procedure can be followed to verify that this part of the circuit is working. Adjust the time delay (E1P) on the Ceiling Sensing board for 300 (3 turns). Remove the voltage at IN1 of the Washout and Output board and press the reset pushbutton on the Ceiling Sensing board. Place the PSS in service (regulator in manual). Put a voltage into IN1 of sufficient magnitude to cause a trip. The PSS should trip to "OFF" approximately 10 seconds after insertion of the signal. Remove the test voltage and reset the Ceiling Sensing board by pressing the reset pushbutton on the faceplate. Place the positive and negative limit dials on 1000 (10 turns) and the time delay dial on zero.

8. On Washout and Output board (LOC 1108) set negative limit dial and positive limit dial on 500 (5 turns). Insert a variable dc voltage into IN1. Monitor

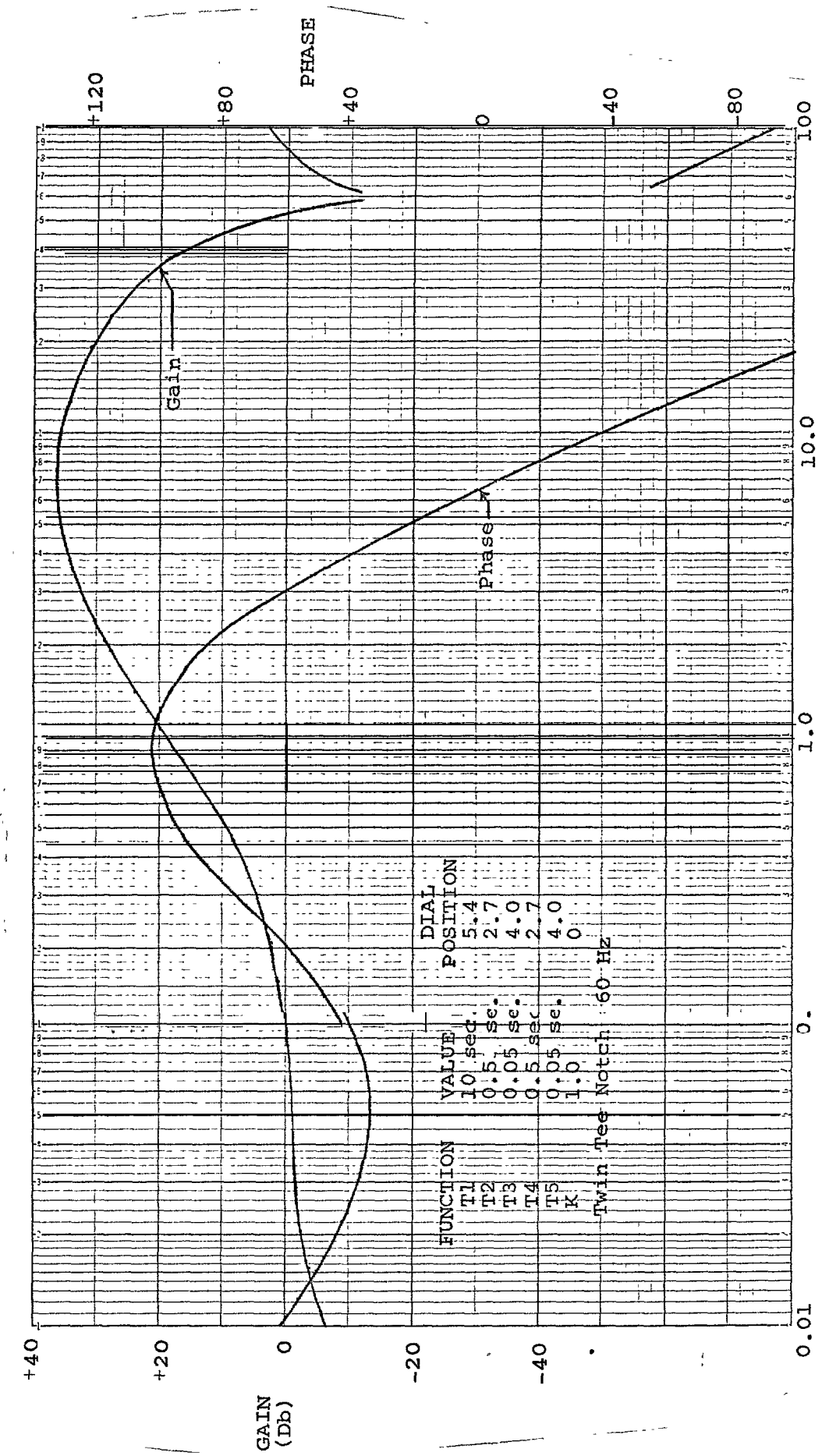


Figure 1
Gain and Phase Vs. Frequency for PSS (Excluding Input Transducer)

[OPERATION (continued)

INITIAL OPERATION (continued)

the output voltage at OUT1 using an oscilloscope or voltmeter. Increase the dc voltage slowly and note at what voltage the output fails to. follow the input voltage (output voltage is limited). This voltage should be 4.75 volts + 0.5 volts. Reverse the voltage and perform the same test. The output voltage should be of the opposite polarity but same magnitude (within ± 0.5 volts). Remove the voltage source and Extender board. Reinsert the Frequency Transducer board. Insure that the PSS is off. Place the regulator in the automatic mode of operation.

9. The adjustment range provided by the PSS can result in extremely high gain over a range of frequencies. For instance, with K set at 100, T2 and T3 set at 1.5 sec. and 0.02 sec. respectively, the gain will be 116 Db (631,000). Although this gain will probably never be used in an actual system, a good check of PSS noise level can be obtained usinq this gain. Insure that the PSS is off. On the Signal Conditioning board-(LOC 1102), set TCT2F and TCT4 for 1000 (10 turns) and TCT3 and TCT5 for zero. Pull the Washout and Output board (LOC 1108). Put the shorting clip on C5S on terminals 2, 3 (GRD). Put the shorting clip on C3S on terminals 2, 3. insert Washout and Output board. Turn gain adjustment to 1000 (10 turns). Remove Frequency Transducer board (LOC 1101). insert Extender board in place of the Frequency Transducer. Short from pin 15 to pin 3 of the Extender board. Using an oscilloscope with a very slow sweep speed (1 sec/cm), determine the peak-to-peak output voltage (measured at OUT 1 of the Washout and Output board). This voltage should be observed for some time, perhaps 10 sec to 1 min. This voltage should be less than 3 volts peak-to-peak. Remove Extender board. Insert Frequency Transducer board in LOC 1101. Remove Washout and Output board. Place C5S shorting clip on terminals 1, 2 (not GRD). Place C3S shorting clip on terminals 1, 2. Return gain adjustment dial to zero (unity gain). insert Washout and Output board. Final settings of time constants and gain can only be made after on-line tests and/or system studi‘es.

10. Check out the following optional circuits if supplied with this equipment.

A. Check out the Voltage Sensing panel located below the printed circuit board rack. Remove the Voltage Deviation Sensing board and the Under-Overvoltage Detection board. Apply 120 VAC to L25, L26 and L27 (ØA, ØB and ØC respectively). Close the isolating switches mounted on the "L" terminal board at L25, L26, and L27. The voltage from LB5-B to LB5-A should be 38 VDC ± 10%. Replace the Voltage Deviation Sensing and Under-Overvoltage Detection boards.

B. Check out the Under-Overvoltage Detection board (LOC 1109). Put the Under-Overvoltage Detection board on an Extender board. Measure the voltage at D5TP. If the voltage is not +5 VDC ± 0.05 VDC, then adjust using D4P. Measure the voltage at D6TP. If the voltage is not -5 VDC ± 0.05 VDC, then adjust using D6P. Measure the voltage at D2TP. If the voltage is not -5 VDC ± 0.05 VDC, then adjust using D2P. At an input vol tage correspond i ng to rated, measure and record the voltage at D1TP. Adjust D1P (input voltage) for zero volts at OUT 1. Place C1S shorting

OPERATION (continued)

INITIAL OPERATION (continued)

clip on the Washout and Output board on terminals 1, 2. The voltage at OUT 2 should be > 2.0 VDC. Adjust the positive and negative limit dial to 0 (0 turns). Change the generator voltage by approximately 1% (or change the voltage at pins 15, 16 by 1% if a dummy voltage source is used). Turn the appropriate limit dial (if voltage was raised, use the positive 1 imit and if lowered, use the negative limit) and determine the setting at which OUT 2 changes to approximately 0.5 volts. The dial setting should be 100 ± 5. When terminal voltage is changed back to nominal, the voltage at OUT 2 should return to > 2.0 VDC.

With the C5S shorting switch on terminals 1, 2 (on the Washout and Output board), insert a +1 VDC signal into IN1 of the Washout and Output board. If noise is present in the output, it may be necessary to pull the Signal Conditioning board. With the gain dial set for zero, the voltage at OUT 1 on the Washout and Output board should be -1 VDC. Change terminal voltage by an amount sufficient to cause the voltage at OUT 2 of the Under-Overvoltage Detection board to change to approximately 0.5 VDC. The voltage at OUT 1 on the Washout and Output board should be reduced to zero volts (± 0.1 VDC) . Change terminal voltage back to nominal and the voitage at OUT 1 on the Washout and Output board should return to -1 VDC after approximately 0.5 seconds. Remove the si gna? from IN1 of the Signal Conditioning board. The positive and negative limit dials should be set to the values specified by the customer. C5S should be left on terminals 2 and 3 and C1S should be left on terminals 1 and 2.

C. Check out the Voltage Deviation Sensing board (LOC 1107). Put the Voltage Deviation Sensing board on the Extender board. Adjust the gain dial for 1000. Place B1S (Signal Conditioning board) on terminals 2, 3. Measure the voltage at H1TP. If this voltage is not -5 VDC ± 0.05 VDC, then adjust using H2P. Place H1S shorting switch on terminals 2, 3 (GRD). With rated generator terminal voltage (or with a source of dc voltage. connected from pin 11 to pin 13 and adjusted for approximately 30 VDC) adjust HIP for zero volts at OUT. Change terminal voltage by 1% (or the dc source by 1%). The voltage at OUT should be 0.26, ± 0.025 VDC. If an external source is used to simulate a terminal voltage signal, the signal should be inserted between pins 11, 12 and pins 13, 14. Open the PT isolation switches at L25, L26 and L27 if an external source is used. Note that the Voltage Deviation Sensing board causes an inversion of the signal. Return the gain dial to zero, remove the external voltage source (if used) and close the PT isolation switches. Remove the Extender board and re-insert the Voltage Deviation Sensing board. B1S should be left on terminals 1 and 2 and H1S should be left on terminals 2 and 3.

Initial Settings

The PSS has been designed to interface with various excitation systems. In order to permit compensation for widely different. systems, adjustments are made using calibrated dials. On the Washout and Output board, set the gain dial to

OPERATION (continued)

Initial Settings (continued)

zero and C3S on 1-2. Insure that the Relay board has the correct rheostat and switch settings. Check the Frequency Transducer board to insure that A1S is on the proper terminals for the desired gain (0.1V/% or 1.0V/%). If the "internal frequency" option is to be used, check that the shorting jumper across LA2-B and LA2-D is removed. If the optional Under-Overvoltage Detection board is being used, insure that C1S has the shorting link on terminals 1, 2. If the optional Voltage Deviation Sensing board is being used, insure that B1S (Signal Conditioning board) has the shorting link on terminals 1, 2. Close the PT isolation switch and the CT isolation switch (if the "internal frequency" option is being used.)

Settings for a particular installation are best obtained by analytical studies and/or on line tests. It is the responsibility of the customer to perform the analytical studies and on line tests necessary to determine proper settings. General Electric Company can provide assistance in performing analytical studies and system tests. Contact your local sales office. A paper, "Concepts of Synchronous Machine Stability as Affected by Excitation Control", by F'. P. deMello and C. Concordia may be used as a reference (68-TP-129-PWR). Other publications which may be used as references are "Test Procedure for Power System Stabilizers" published by the Western Systems Coordinating Council and an IEEE publication, "Guide for the Identifi-cation, Testing and Evaluation of the Dynamic Performance of Excitation Control Systems" paragraph 6.03 (P421, 1977).

On-Line Tests

Power System Stabilizers are employed to provide an additional input to the regulator to improve power system dynamic performance. They have been employed to solve two separate but related problems; local machine-system oscillations in the 1-2 Hz range and inter-area oscillations ≈ 0.5 Hz. The fundamental function of the signal conditioning network of the PSS is to compensate for the phase lags of the system being controlled. For best contribution to system damping, the gain of the PSS should be maximized under the most sensitive conditions (usually at or near full load). The desired stabilizing signal has as its reference the base system frequency which is slowly changing all the time. The washout function serves to minimize the effect of overall system speed changes on machine terminal vol tage.

The PSS 'has adjustable limits on the output signal so that the PSS will not completely override voltage regulator action. The output voltage limits are independently adjustable but should be set identically in order to prevent non-symmetrical limiting which might bias the voltage regulator. Since the ceiling sensing function occurs after the adjustable output limit stage, the output voltage limits should be set at a higher voltage than the ceiling sensing limits.

In order to facilitate on line frequency response testing of the excitation system, IN1 is provided on the Washout and Output board. With the shorting clip on C5S on terminals 1, 2, a drive signal can be inserted into IN1 which will produce a calibrated terminal voltage change through the regulator. For normal

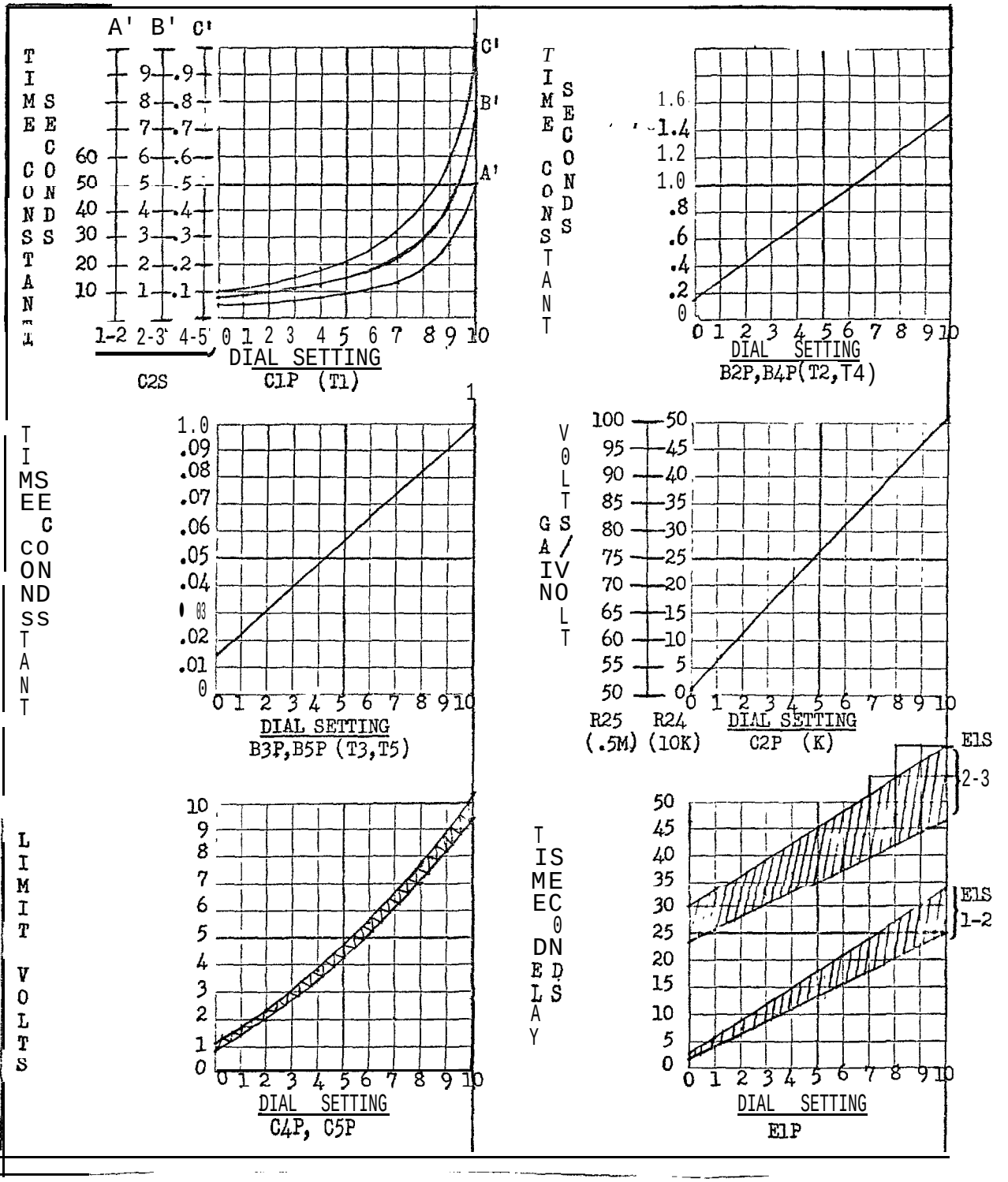


Figure 2
Parameter Value Vs Dial Setting

OPERATION (continued)

On-Line Tests (continued)

operation, the shorting clip on C5S should be on terminals 2, 3 (GRD) and the shorti'ng clip on C1S should be on terminals 2, 3.

Figure 2 is provided so that dial setti ngs can be chosen to real i ze the des i red time constants and limit settings.

PRINCIPLES OF OPERATION

GENERAL

The Power System Stabilizer is used to provide an additional input to the voltage regulator which, under certain system conditions, can improve power system dynamic stability. A signal is obtained from terminal voltage (or optionally terminal voltage added vectorially to line current).' This signal which is frequency modulated with the carrier frequency being 60 Hz and the modulation frequency being equal to the frequency changes of terminal voltage, This frequency modulated signal is passed through a bandpass filter and frequency transducer to produce an output signal proportional to the magnitude of the deviation in frequency with a frequency equal to the deviation frequency. This demodulated signal is then passed through two stages of lead-lag compensation to provide phase compensation for the system, a twin-tee notch fi 1 ter tuned to 60 Hz, a washout stage to remove drift and slow system frequency changes and an adjustable gain stage to maximize the effect of the PSS. Limits are provided on the output stage so that during large system disturbances, the PSS will not interfere with voltage regulator action. A relay is provided for placing the PSS in service (and also removal from service). Interfacing is provided to properly scale the PSS output signal so that 1 volt from the output stage produces a 1% change in terminal voltage. Protective circuits are provided to detect a failure which causes the output to go to ceiling (Ceiling Sensing board) and optionally a failure or system condition-which causes terminal voltage to deviate by more than a preset value from nominal (Under-Overvoltage Detection board). Ten-turn rheostats with ca? ibrated dials are furnished for al? customer adjustments. Figure 6 includes an overall block diagram of the PSS. Individual board and pane? elementary diagrams are furnished with the equipment. it will be necessary to have these for much of the following explanations.

POWER SUPPLY SYSTEM

The pane? mounted PSS includes both the ± 15 VDC and 24 VDC power supplies. The 115 VAC, 60 Hz power should be connected to L21 and L22. The S1T and S2T transform the input voltage (and provide isolation) and then the voltage is rectified by a single phase full-wave bridge, f i 1 tered to remove the dc component, 1 imi ted by Zener diode action and then fed to series pass regulators. Fuses R1F and R2F, SCR's R1CR and R2CR, along with associated sensing and firing circuits form a "crow-bar" circuit which wi 11 protect sensitive semiconductor components from overvoltage by blowing a fuse and thus removing the excess voltage. IC voltage regulators (G1A, G2A) are used to control the output voltage. The G2P and G4P

PRINCIPLES OF OPERATION (continued)

POWER SUPPLY SYSTEM (continued)

mounted on the printed circuit board provide adjustment of the output voltage. The G1R, G1P, G2R, G3R provide a current limit feature on the +15 volt regulator (a similar circuit is found on the -15 volt regulator). The current limit is of the foldback type, i.e., the maximum output current is reduced as the load on the regulator is increased above the full-load value. This insures that critical components in the regulator (series pass transistors) will not be damaged by a short circuit. The current limit can be adjusted using G1P and G5P. The positive limit is factory set at 1 Amp and the negative limit at 750 Ma (these are not the short circuit current' values) . Integrated circuits J1A and J2A provide voltage sensing for the "crow-bar" circuit. The J1P and J2P are factory adjusted to cause the "crow-bar" circuit to operate at a supply voltage of 17 volts. The S1T and S2T have taps which will permit variations in the input voltage from the nominal 115 VAC value. (Refer to the diagram for the correct tap for a given input voltage.) In order to reduce the noise level to as low a value as possible, the + 15 volt power supply is very tightly regulated and heavily filtered. The regulation-is 0.1% and the ripple on the power supply should be ≤ 5 Mv PP. The 24 volt supply does not require tight regulation since it is used to drive relays and energize lights. The S3T is used to isolate the 24 volt supply from the $\pm$ 15 volt supply and to transform the input voltage to a lower voltage. This voltage is rectified by a full-wave bridge and then filtered. Regulation is provided by a Zener diode (S1Z). The 24 volt power supply is ungrounded while the $\pm$ 15 volt supplies are grounded through a parallel resistor capacitor combination. The high impedance ground on the $\pm$ 15 volt supplies allows the PSS to be used with both grounded and ungrounded systems without ground loop problems.

INTERNAL FREQUENCY PANEL

The Internal Frequency panel contains potential transformers which accept a three-phase voltage proportional to generator terminal voltage (115 VAC) and generate a single phase voltage ($\approx$ 3.5 VAC) in phase with phase B line-to-neutral voltage. Taps are provided on the secondary of L1T and L2T to provide 2 VAC on each secondary with an input voltage which varies from 110 VAC to 120 VAC. The L1T and L2T are connected open delta with the polarity of L1T reversed on the secondary. This gives the proper phase shift so that when L1T voltage is added to L2T voltage, a voltage in phase with B phase line-to-neutral voltage is developed.

A reactor is provided which develops a voltage proportional to line B current (5 Amps at 1 per unit load) and phase shifted from line B current. If it is desired to use the so called "internal frequency" option, a shorting jumper is removed from the secondary of LIX and the CT shorting switch is closed. This gives a sensing voltage equal to Vt + jIX the frequency of which, under most conditions, approximates the speed of the rotor. The reactance should be equal to X_Q of the generator. Figure 3 shows the phasor diagrams for generating a voltage having the so-called "internal frequency".

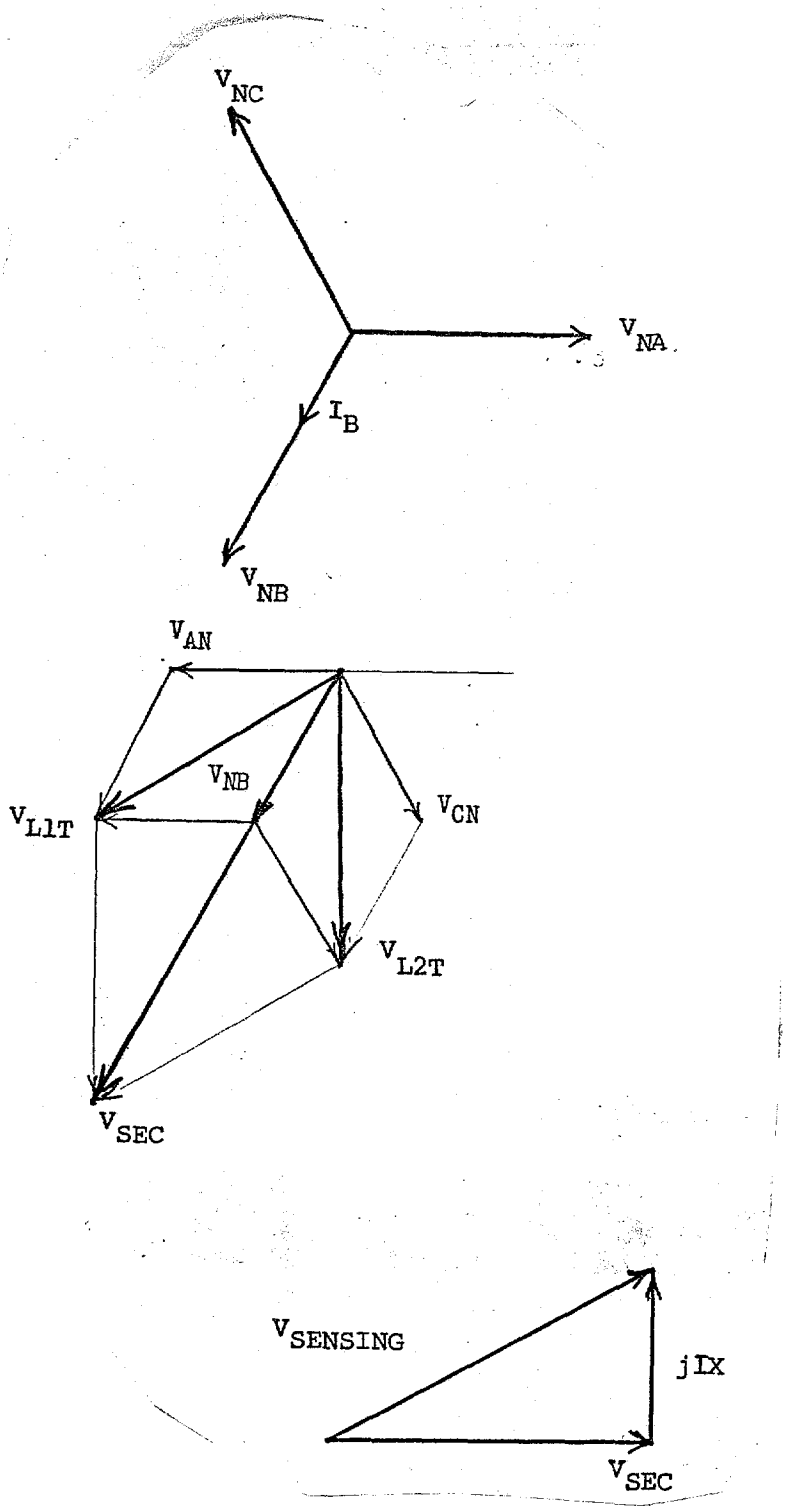


Figure 3

/PRINCIPLES OF OPERATION (continued)!

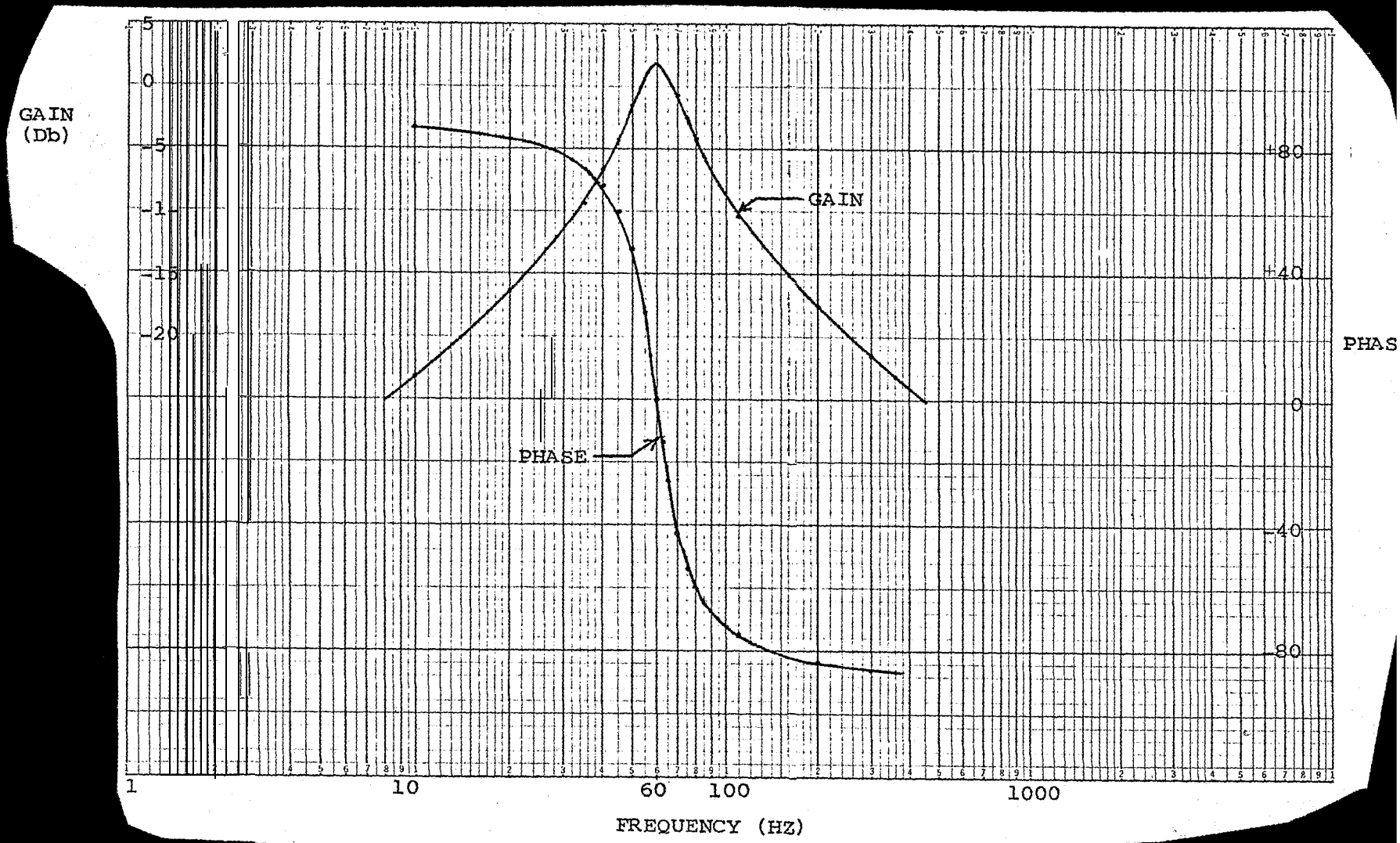
60 HERTZ BANDPASS FILTER BOARD

Tests on a PSS using a frequency transducer to demodulate a terminal voltage signal indicate that excessive power system noise can render the PSS inoperative. By providing an input filter to remove the higher frequency noise components, acceptable noise levels have been obtained. The bandpass filter is positioned to filter the combined voltage and current signal (if a current signal is used). A single operational amplifier (with associated resistors and capacitors) is used to construct the bandpass filter. This realization gives no inversion of the signal and offers a relatively high degree of insensitivity to component drift. Center frequency is determined by 1% capacitors and 1% resistors which give a ± 2% tolerance of center frequency in the worst case. The "Q" of the filter (defined as $Q = fo/f_2-f_1$) where fo is the center frequency, f2 is the upper'frequency at which the amplitude is down 3Db from the amplitude at center frequency and fl is the lower frequency at which the amplitude is down 3Db from the amplitude at center frequency) is set for 3.0. This has provided adequate noise reduction while providing only 3 degrees of phase lag at 0.5 Hz and about 9 degrees of phase lag at 1.5 Hz. Fine adjustment of the "Q" is provided by P1 which is a screwdriver adjustment through a hole in the faceplate. The gain of the filter is proportional to "Q" and so 1R and 2R form a voltage divider circuit to give an overall gain at center frequency of approximately 1.25. The phase shift through the filter at center frequency is zero. This provides an accurate method for determining center frequency. Figure 4 shows the frequency response of the filter.

FREQUENCY TRANSDUCER BOARD

The Frequency Transducer (LOC 1101) produces a signal proportional to the frequency of an input voltage. A single phase voltage proportional to phase B voltage (or synthesized generator internal voltage) is fed to a crossover detector (A1A and associated circuitry) which produces square waves having constant amplitude and a period equal to the period of the incoming signal. Hysteresis is provided by A2Q, A3Q, A3D, A4D, A5D, A6D and A1Z *to* prevent chattering of the crossover detector (42 Mv of hysteresis). The A2Z is used to provide a precise amplitude signal to integrator A2A. The A5Q is a constant current source to further stabilize the voltage on A2Z. The A6Q is used to force the voltage across A2Z to zero during the positive half-cycle of input voltage.

On the negative half-cycle of input voltage, A2Z develops a constant voltage and this is integrated by A2A. At the beginning of the positive half-cycle, A2Z will have zero volts and a one-shot will trigger a sample and hold device to sample the peak of the integrator ou-tput (proportional to the time of a half-cycle) and hold this value until the next negative half-cycle discharges the integrating capacitors and resets the integrator. The output of.the sample and hold device is fed to A3A which also has a bias signal of the opposite polarity to give zero volts at rated frequency. A jog pushbutton is provided which causes the output of the Frequency Transducer to change (-) 0.005 volts when pressed. The A4A develops adequate gain for either 1 volt/% change in frequency or 0.1 volt/% change in frequency selected by a jumper A1S mounted on the printed circuit board.


GAIN
(Db)
5
0
-5
-1
-15
-20
+80
+40
0
-40
-80
PHAS
GAIN
PHASE
1
10
60
100
1000
FREQUENCY (HZ)

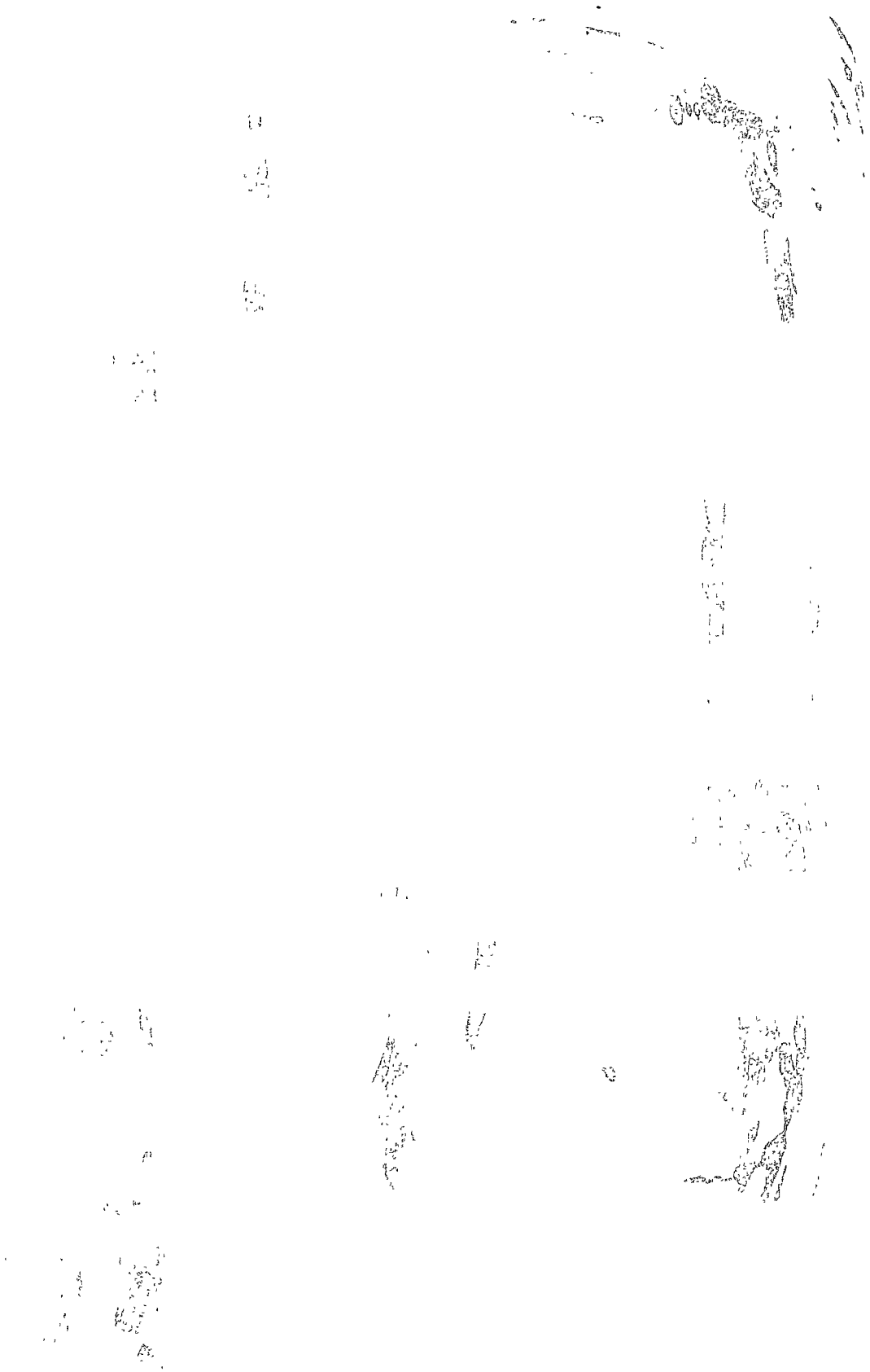


Figure 4
Gain and Phase Vs Frequency for 60 Hz Bandpass Filter

PRINCIPLES OF OPERATION (continued)

SIGNAL CONDITIONING BOARD

The Signal Conditioning board consists of two lead-lag stages and a twin-tee notch filter. In addition to the signal from the Frequency Transducer, an extra input with separate lead time constant adjustment is provided for a voltage input (IN2). The B1P provides adjustment of the first stage lead time constant which is proportional to B1C, B1R and the setting of B1P. The B3P provides adjustment of the lag time constant which is proportional to B9R, B4C and the setting of B3P. The dc gain of the stage is B9R/B1R which is 1.0. The B3R, B3C, and the equivalent resistance of B3P cause extraneous lead and lag time constants which have been minimized but may be significant when performing calculations at torsional frequencies. The second lead-lag stage is identical to the first except for the addition of an output voltage 1 imit circuit formed by B1Z, B2Z, B5D, B6D and a drive circuit formed by B1Q, B21R, B22R, B3D and B4D to increase the output current capability of B2A. A twin-tee notch filter is placed between the f i rs t and second 1 ead- lag stages to remove 60 Hz. The twi n-tee is factory adjusted to produce greater than 50 Db attenuation at the notch frequency . The notch frequency is given by

$$f = \frac{1}{2\pi(B1_1R+B4P)\overline{(B16C)}}$$

WASHO'UT AND OUTPUT BOARD

The Washout and Output board contains a washout function, a gain function and a power ampl i f icat ion function. Operational amplifier C1A is a chopper stabilized device fbr low offset. The C1A is basically a derivative circuit- formed by C1C and the feedback resistor (C2R, C3R or C4R). The C2C is added to make the derivative circuit only effective for low frequencies and C1P is used to provide adjustment of the effective feedback resistance and thus control the time constant. Coarse adjustment of the time constant is available through C2S. The C9D, C10D, Z1C, Z2C, C31R and C32R form a clamp to limit the output voltage of the washout stage. An extra input is available at IN1 which bypasses the washout stage. This allows insertion of a signal into the voltage regulator for calibration of the PSS or for frequency response testing of the system. The C5S is used to ground IN1 during normal use. Changing C5S shorting clip to terminals 1, 2 allows IN1 to be used for driving the regulator. The C2A is used as an adjustable gain stage with the gain being determined by C3S and C2P. The C3S is used to select a range of gain from 1 *to* 50 or 50 to 100. An adjustable limit is provided on the output of C2A by C1Q, C2Q, C15R, C16R, C1D, C2D, C4P, C5P, C13R and C14R. The output of C2A is used as the input signal for Ceiling Sensing board through pins 13, 14. The C3Q and C4Q are junction FET's used as a static switch. The IN3 can be connected to an optional Under-Overvoltage Detection board which serves to disconnect the PSS whenever terininal voltage deviates by an adjustable amount. In normal operation, C1S should have the shorting clip between terminals 2, 3 (OUT). The C3Q will be conducting and C4Q will be nonconducting. The C3A is connected to a push-pull transistor stage (C7Q, C8Q, C9Q, C10Q) in order to increase the output current capability to ± 100 Ma. The C3A and the push-pull stage are connected as a unity gain voltage follower.

PRINCIPLES OF OPERATION (continued)

RELAY BOARD

The Relay board is used- to control insertion of the PSS into the regulator, provide the proper interfacing with different voltage regulators and provide interfacing with customer equipment. Meter Ml is mounted on the faceplate and can be used to insure that the PSS output is zero before insertion into the voltage regulator. OFF and ON pushbuttons with lights are mounted on the faceplate and can be used to insert or remove the PSS from service. Lights mounted in the pushbutton indicate the status of the PSS. A line breaker contact (52G) should be used as a permissive contact so that the PSS is inoperative off line. The pushbuttons have momentary contacts and so F93 is sealed in by an FAR and F93 contact. The F93A is an auxiliary relay used to increase the number of contacts of F93. FAR is controlled by the Ceiling Sensing board.

The FIR, F2R, and F1P are used to properly scale the PSS signal for insertion into the' regulator. The F2P is used to dummy load the regulator so that a bump in excitation will not occur whenever the PSS is placed in service. (Refer to the diagram for the proper settings of F1P and F2P and switches F1S, F2S, F3S, F4S.)

CEILING SENSING BOARD

The Ceiling Sensing board is used to detect a failure in the-PSS which causes the output to be greater than a specified value for an adjustable length of time. Once the condition exists, the PSS cannot be restored to service until a reset pushbutton is depressed on the Ceiling Sensing board.

The E1A and E2A operate as comparators with E2P and E3P used to adjust the level at which a failure is detected. If the input voltage is less than the positive limit (E2P) and more positive than the negative limit (E3P) then the voltage at E6TP is at a minimum. The voltage at E5TP is passed through an inverter in E1L and then a nand operation is performed on the signal from E6TP and the inverted signal from E5TP. A nand gate has its output in the high state unless all inputs are in the high state.

In this case, both inputs are in the high state so its output is in the low state. The output of this nand gate is passed through two inverters (no change in state) and is also used to trigger a one-shot (E3L). With a ceiling condition not detected, the one-shot trigger input will stay in the low state'and the output (the complemented output is used in this case) stays in the high state. The inputs to the first nand gate of E2L are a low and a high, which means the output will be in the high state. Two of the nand gates are used in a cross-coupled fashion to allow latching whenever a ceiling condition is detected. The E8C is provided to insure that the cross-coupled gates will go to the proper state upon energization. With no ceiling condition detected, the inverter output of E2L will be in the low state turning on transistors E2Q and E3Q and energizing relay FAR on the Relay board.

If a ceiling condition.exists (whether positive or negative), the second nand gate of E1L will have its input states changed to one low and one high. The output of this nand gate will go to the high state which triggers the one-shot (E3L)

PRINCIPLES OF OPERATION (continued)!

CEILING SENSING BOARD (continued)

and changes the input on the first nand gate of E2L to a high state. Nothing happens, however, because the output of the one-shot (E3L) is in the low state and remains there for a period of time determined by E7C, E1P and E8R or E9R (depending on the setting of E1S). After the timing interval, the output of the one-shot (E3L) goes to the high state and the output of the first nand gate of E2L goes to the low state. The output of the cross-coupled gates of E2L goes to the low state and will remain low even if the ceiling condition is cleared. The inverter output of E2L changes to the high state which turns off E2Q and E3Q and drops out relay FAR on the Relay board, giving an alarm. To clear the alarm and place the PSS back in service, it is necessary to press the reset pushbutton (E1PB). This changes the input to one of the cross-coupled gates from a high to a low state which forces the output of the inverter to be in a low state. The cross-coupled gates latch in this state, the relay FAR is picked up and the alarm is cleared. Table 1 illustrates the logic levels at the various board test points for both normal and abnormal input signals.

TABLE 1

Input Vo 1 tage	E5TP	E6TP	E 7TP	E8TP Init	E8TP Time Delay	E9TP	11TP and "Out' Init	11TP and "Out' Time Delay
Normal	0	1	0	1		0	0	
High Positive	1	1	1	0	1	1	0	1
High Negative	0	0	1	0	1	1	0	1

Note: A zero is ≤ 0.6 volts
A one is ≥ 2 volts

Logic Levels for Ceiling Sensing Board

OPTIONAL EQUIPMENT

Under-Overvoltage Detection Board (Optional)

The Under-Overvoltage Detection board (LOC 1109) is used to remove the PSS from service whenever generator terminal voltage deviates by a preset amount from nominal. The D2P is used to establish a reference for the input voltage signal and insure proper scal ing. The D1P is used to null out the reference voltage at rated terminal voltage so that a change from rated is detected. A faceplate mounted test point (IN2) is provided for insertion of a test signal. The D2A and D3A are used as comparators with D3P and D4P establishing a variable reference point. The D4P and D6P provide proper scaling so that D3P and D4P dial setting

PRINCIPLES OF OPERATION (continued)!

OPTIONAL EQUIPMENT (continued)

(0 to 10) corresponds to percent voltage deviation from nominal. The output normally stays at > 2.0 VDC with no undervoltage or overvoltage detected. When an undervoltage or overvoltage is detected, the output goes to ≈ 0.5 VDC. If terminal voltage changes back *to* a value less than the limits, the output changes back to > 2 VDC after 1/2 sec time delay. The output of the Under-Overvoltage Detection board is used to control a static switch on the Washout and Output board. If the Under-Overvoltage Detection board is to be used, then the shorting link on CIS must be placed between terminals 1 and 2. Nominal input voltage is 30 VDC at rated terminal voltage.

Voltage Deviation Sensing Board (Optional)

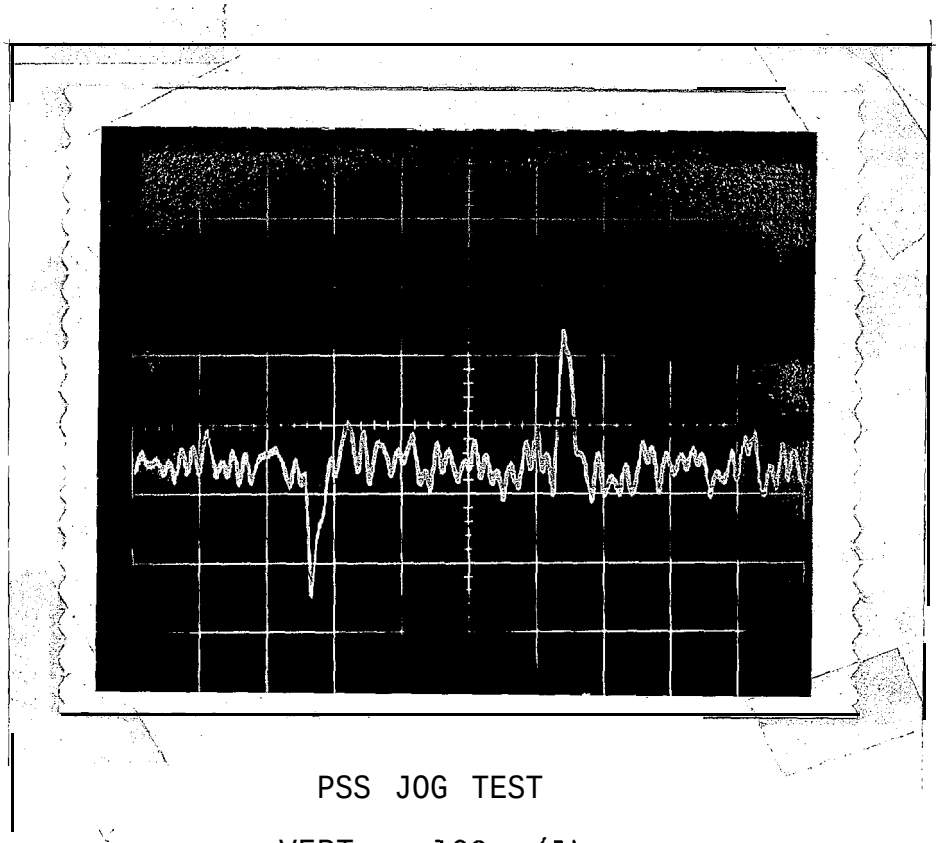
The Voltage Deviation Sensing board (LOC 1107) is used to provide an input to the PSS proportional to the deviation of terminal voltage from rated. The input signal is obtained from the Voltage Sensing panel and is nominally 30 VDC at rated generator terminal voltage. The H2P is used to provide a reference voltage and to allow proper scaling of the input signal. The H1P is used to null out the reference voltage at rated terminal voltage. A faceplate mounted test point (IN2) allows insertion of a signal for test purposes. Gain adjustment is provided for a range of 0.2 to 5.2 volts per volt or 0.01 to 0.26 per unit/per unit. If the Voltage Deviation Sensing board is to be used, the shorting link on B1S (Signal Conditioning board) must be on terminals 2 and 3, (not GRD) .

MAINTENANCE

The PSS is a static device having no moving parts except for relays and instruments, therefore, little maintenance should be required. If excessive vibration is present, all screw connections should be checked periodically. The Frequency Transducer null adjustment should be checked periodically and readjusted if necessary. A pushbutton is furnished on the Frequency Transducer which puts a small step into the Frequency Transducer output. The step is equivalent to approximately 0.005% change in frequency. This change is so small that system noise may mask the effect on some systems.- If a response can be observed in the output of the Washout and Output board when pressing the pushbutton, record for future reference. A quick check on the operating status of the PSS can then be made by pressing the pushbutton and comparing the output with the recorded output. The following photograph (Figure 5) shows the output of the Washout and Output board (T_2 = T4 = 0.5 sec, T3 = T5 = 0.05 sec, T1 = 10 sec and K = 1.0) when the pushbutton is pressed and then released.

TROUBLESHOOTING I

1. Carefully check all wiring if problems are experienced at start-up.
2. If no response is obtained while making the frequency response test:
   a. Check that the power switch on the Power Supply board is on.



PSS JOG TEST

VERT. = 100mv/div.
HORIZ. = 200 msec/div.

Figure 5

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